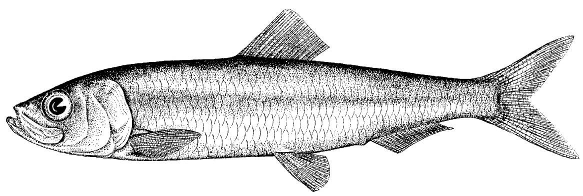
Pacific Herring preliminary data summary for West Coast of Vancouver Island 2025

Jaclyn Cleary* Matthew Grinnell†

August 26, 2025



Pacific Herring (*Clupea pallasii*). Image credit: Fisheries and Oceans Canada.

Disclaimer This report contains preliminary data collected for Pacific Herring in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). These data may differ from data used and presented in the final stock assessment.

---

1 Context

Pacific Herring (*Clupea pallasii*) in British Columbia are assessed as 5 major and 2 minor stock assessment regions (SARs), and data are collected and summarized on this scale (Table 1, Figure 1). Note that formal stock assessments are only done for major SARs. The Pacific Herring data collection program includes fishery-dependent and -independent data from 1951 to 2025. This includes annual time series of commercial catch data, biological samples (providing information on proportion-at-age and weight-at-age), and spawn index data conducted using a combination of surface and SCUBA surveys. In some areas, industry- and/or First Nations-operated in-season soundings programs are also conducted, and this information is used by resource managers, First Nations, and

*DFO Science, Pacific Biological Station (email: Jaclyn.Cleary@dfo-mpo.gc.ca).
†DFO Science, Pacific Biological Station (email: Matthew.Grinnell@dfo-mpo.gc.ca).

stakeholders to locate fish and identify areas of high and low Pacific Herring biomass to plan harvesting activities. In-season acoustic soundings are not used by stock assessment to inform the estimation of spawning biomass.

The following is a description of data collected for Pacific Herring in 2025 in the West Coast of Vancouver Island major SAR (Figure 2). Data collected outside the SAR boundary are not included in this summary, and are not used for the purposes of stock assessment. Although we summarise data at the scale of the SAR for stock assessments, we summarise data at finer spatial scales in this report: Locations are nested within Sections, Sections are nested within Statistical Areas, and Statistical Areas are nested within SARs (Table 2). Note that we refer to 'year' instead of 'herring season' in this report; therefore 2025 refers to the 2024/2025 Pacific Herring season.

2 Data collection programs

Biological samples were collected by the seine charter vessel *Western Investor* for 17 days from February 19th to March 8th. The primary purpose of the test charter vessel was to collect biological samples from main bodies of herring from Statistical Areas 23, 24, and 25. Nearshore herring samples were collected by the Nuu-chah-nulth staff as part of a sampling program and on-going collaboration between WCVI First Nations and DFO. These nearshore biological samples were collected from spawning aggregations using cast nets. WCVI First Nations spawn reconnaissance charters reported spawn activity in all three areas. Herring spawn locations were primarily identified with fixed-wing overflights conducted by DFO Resource Management area staff. Eleven flights were conducted this season all in March.

One dive charter vessel and a shorebased crew operated in WCVI:

- The *Pachena No. 1* surveyed 19 days from March 9th to March 27th, and
- The shorebased dive surveyed 15 days from February 17th to March 24th.

3 Catch and biological samples

In the 1950s and 1960s, the reduction fishery dominated Pacific Herring catch; starting in the 1970s, catch has been predominantly from roe seine and gillnet fisheries. The reduction fishery is different from current fisheries in several ways. First, the reduction fishery caught Pacific Herring of all ages, whereas current fisheries target spawning (i.e., mature) fish. Thus, reduction fisheries included an unknown number of age-1 fish which are not typically caught in current fisheries. Second, the reduction fishery has some uncertainty regarding the quantity and location of catch; however catches have been resolved to SAR and Statistical Area using fish slips as best as possible. For the roe gillnet fishery, all Pacific Herring catch has been validated by a dockside monitoring program since 1998; the catch validation program started in 1999 for the roe seine fishery. Finally, the reduction fishery operated during the summer and winter months, whereas roe fisheries typically target spawning fish between February and April.

Landed commercial catch of Pacific Herring by year and fishery is shown in Table 3 and Figure 3. Total harvested spawn-on-kelp (SOK) in 2025 in the West Coast of Vancouver Island major SAR is shown in Table 4; we also calculate the estimated spawning biomass associated with SOK harvest. See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. Incidental mortality from fisheries and aquaculture activities is shown in Figure 4.

In 2025, 33 Pacific Herring biological samples were collected and processed for the West Coast of Vancouver Island major SAR (Table 5, Table 6). The nearshore sampling program is a multi-year pilot study using cast nets (Figure 10, Figure 11, and Figure 12). Differences between biological data collected from two sampling protocols regarding the number-, proportion-, weight-, and length-at-age for Pacific Herring in 2025 in the West Coast of Vancouver Island major SAR are shown in Table 7, Table 8, Table 9, and Table 10, respectively. Summaries of data collected by the nearshore sampling program are shown in Table 11, Table 12, Table 13, and Table 14. We compare length-, weight, and proportion-at-age as well as length-weight relationships for nearshore and seine test samples in Figure 13, Figure 14, Figure 15, and Figure 16, respectively. Note that only biological data from the seine samples are used for the purposes of stock assessment. The locations in which the biological samples were collected are presented in Figure 5. Included herein are biological summaries of observed proportion-, number-, weight-, and length-at-age (Figure 6, Table 15, and Figure 7, respectively). We also show the percent change in weight and length for age-3 and age-6 fish (Figure 8 & Figure 9, respectively). Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. Only representative biological samples are included, where 'representative' indicates whether the Pacific Herring sample in the set accurately reflects the larger Pacific Herring school.

3.1 Nearshore genetics project

DFO initiated a nearshore genetics project in 2022 to better understand the spatial and temporal structure of Pacific Herring stocks in British Columbia. We do this by collecting genetic and biological data from actively spawning Pacific Herring (*Clupea pallasii*). Then we collect biological data (e.g., length, weight, age, sex) and genetic data from the fish. Finally, we analyse the biological and genetic data to better understand Pacific Herring. Genetic samples were collected in the West Coast of Vancouver Island major SAR in 2024 (Figure 17). Results from the nearshore genetics project will be presented in a separate document when they are available.

4 Spawn survey data

Pacific Herring spawn surveys were conducted at 54 individual locations in 2025 in the West Coast of Vancouver Island major SAR (Table 16, and Figure 18). A summary of spawn from the last decade (2015 to 2024) is shown in Figure 19. Figure 20 shows spawn start date by decade and Statistical Area. Spawn surveys are conducted to estimate the spawn length, width, number of egg layers, and substrate type, and these data are used to estimate the index of spawning biomass (i.e., the spawn index; Figure 21, Figure 22, Figure 23, Figure 24, Table 17, and Figure 25). See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Therefore, these data do not represent model estimates of spawning biomass, and are considered the minimum observed spawning biomass derived from egg counts. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

Some Pacific Herring Sections contribute more than others to the total spawn index, and the percentage contributed by Section varies yearly (Figure 24b, Figure 26). For example, in 2025, Section 232 contributed the most to the spawn index (53%). As with Sections, some Statistical Areas contribute more than others to the total spawn index (Figure 24c, Figure 27). An animation shows the spawn index by spawn survey location from 1951 to 2025 (Figure 28).

5 First Nations observations

The following observations were contributed by representatives of First Nations communities. These observations provide context and perspectives to this data report. In some cases we make minor edits for clarity and brevity, but we do not change the intent or substance of responses.

5.1 Nuu-chah-nulth

5.1.1 Area 23

There was mixed success with Siihmuu or Spawn on Boughs (SOB) harvests in 2025 but overall, SOB harvest success was comparable to 2024, with the exception of the spawn around Nettle Island. For the most part, spawning duration in all areas was short except for Nettle Island where spawning occurred over three days. As in previous years, the short duration of most spawns necessitated harvesters be on-site at the time of spawning to get boughs in the spawn. There was very limited success for SOB in all areas except Nettle Island. One harvester from Nettle Island reported boughs covered with up to 15 layers.

5.1.2 Area 24

As in previous years, spawning activity in Area 24 began in early January in Hesquiaht Harbour and spread to several other areas in Clayoquot Sound in February and March. There was a noticeable absence of spawn in the areas around Tofino and Opisaht compared to 2024. Both Hesquiaht and Ahousaht had very successful SOB harvests from their territories. SOK trimmings from the five Nations open SOK ponds in Sidney Inlet were provided to community members.

5.1.3 Area 25

Only MMFN was successful with their kaqmis harvest. MMFN set all their trees in Santa Gertrudis Cove with all boughs having sufficient layers for harvesting. Spawning duration was up to two days in lower Nootka Sound, facilitating successful harvests. Nuchatlaht set trees in Rosa Harbour without success. Most spawns in the Nuchatlaht area were short in duration.

6 General observations

The following observations were reported by DFO Resource Management staff and DFO Science staff. These observations provide additional context to this data report.

- Active spawn timing was widespread, with active spawn reported or observed between January 9th and April 27th.
- Vessel and shore-based dive surveys were performed on all observed spawns except for early Hesquiat Harbour, Bartlett Island, Whaler Islet, outside of Nootka Island, Quatsino Sound, late Ucluelet Harbour, and late Nootka Sound.
- Increased FSC harvest (anecdotally) and five Nations right-based sale fishery for SOK opened for the first time.

7 Tables

Table 1. Pacific Herring stock assessment regions (SARs) in British Columbia.

Name	Code	Type
Haida Gwaii	HG	Major
Prince Rupert District	PRD	Major
Central Coast	CC	Major
Strait of Georgia	SoG	Major
West Coast of Vancouver Island	WCVI	Major
Area 27	A27	Minor
Area 2 West	A2W	Minor

Table 2. Statistical Areas and Sections for Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR).

Region	Statistical Area	Section
West Coast of Vancouver Island	23	230
West Coast of Vancouver Island	23	231
West Coast of Vancouver Island	23	232
West Coast of Vancouver Island	23	233
West Coast of Vancouver Island	23	239
West Coast of Vancouver Island	24	240
West Coast of Vancouver Island	24	241
West Coast of Vancouver Island	24	242
West Coast of Vancouver Island	24	243
West Coast of Vancouver Island	24	244
West Coast of Vancouver Island	24	245
West Coast of Vancouver Island	24	249
West Coast of Vancouver Island	25	250
West Coast of Vancouver Island	25	251
West Coast of Vancouver Island	25	252
West Coast of Vancouver Island	25	253
West Coast of Vancouver Island	25	259

Table 3. Total landed commercial catch of Pacific Herring in metric tonnes (t) by gear type in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). Legend: 'Other' represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; 'RoeSN' represents the roe seine fishery; and 'RoeGN' represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Note: data may be withheld due to privacy concerns (WP).

Gear	Catch (t)
Other	0
RoeSN	0
RoeGN	0

Table 4. Total harvested Pacific Herring spawn-on-kelp (SOK) in pounds (lb), and the associated estimate of spawning biomass in metric tonnes (t) from 2015 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. Note: data may be withheld due to privacy concerns (WP).

Year	Harvest (lb)	Spawning biomass (t)
2015	0	0
2016	0	0
2017	0	0
2018	0	0
2019	0	0
2020	0	0
2021	0	0
2022	0	0
2023	0	0
2024	0	0
2025	30,235	45

Table 5. Number of Pacific Herring biological samples processed from 2015 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). Each sample is approximately 100 fish. Note: Nearshore samples are not used in stock assessments.

	Number of samples			
Year	Commercial	Test	Nearshore	Total
2015	0	17	3	20
2016	0	15	10	25
2017	0	12	7	19
2018	0	20	22	42
2019	0	14	11	25
2020	0	12	7	19
2021	0	13	9	22
2022	0	10	19	29
2023	0	9	21	30
2024	0	15	23	38
2025	0	16	17	33

Table 6. Number and type of Pacific Herring biological samples processed in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). Each sample is approximately 100 fish.

Type	Gear	Use	Number of samples
Test	Other	Nearshore	17
Test	Seine	Test fishery	16

Table 7. Observed number-at-age of Pacific Herring by sample type in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Number-at-age								
Sample type	2	3	4	5	6	7	8	9	10
Nearshore	53	210	322	402	173	84	7	3	1
Seine test	47	258	314	487	177	106	21	1	NA
Total	100	468	636	889	350	190	28	4	1

Table 8. Observed proportion-at-age of Pacific Herring by sample type in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

	Proportion-at-age								
Sample type	2	3	4	5	6	7	8	9	10
Nearshore	0.042	0.167	0.257	0.320	0.138	0.067	0.006	0.002	0.001
Seine test	0.033	0.183	0.223	0.345	0.125	0.075	0.015	0.001	NA
Total	0.038	0.176	0.239	0.333	0.131	0.071	0.011	0.002	0.000

Table 9. Observed mean weight-at-age in grams (g) of Pacific Herring by sample type in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

	Mean weight-at-age (g)								
Sample type	2	3	4	5	6	7	8	9	10
Nearshore	58	71	92	107	118	122	141	128	119
Seine test	64	81	100	120	134	133	138	141	NA
Total	61	77	96	114	126	128	139	131	119

Table 10. Observed mean length-at-age in millimetres (mm) of Pacific Herring by sample type in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

	Mean length-at-age (mm)								
Sample type	2	3	4	5	6	7	8	9	10
Nearshore	155	167	181	191	196	200	205	204	204
Seine test	157	171	183	193	199	200	201	205	NA
Total	156	169	182	192	198	200	202	204	204

Table 11. Observed number-at-age of Pacific Herring for nearshore samples from 2015 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Number-at-age								
Year	2	3	4	5	6	7	8	9	10
2015	67	64	130	10	3	3	0	0	0
2016	32	537	161	122	13	3	2	0	0
2017	43	56	355	89	64	4	0	0	0
2018	171	203	142	1157	196	105	18	0	1
2019	101	616	82	37	106	15	6	0	0
2020	168	161	179	28	8	34	6	2	2
2021	47	479	146	90	11	5	8	1	1
2022	138	465	656	160	136	16	7	11	4
2023	23	798	484	384	105	52	8	1	3
2024	24	480	885	326	218	62	8	2	3
2025	53	210	322	402	173	84	7	3	1

Table 12. Observed proportion-at-age of Pacific Herring for nearshore samples from 2015 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Proportion-at-age								
Year	2	3	4	5	6	7	8	9	10
2015	0.242	0.231	0.469	0.036	0.011	0.011	0.000	0.000	0.000
2016	0.037	0.617	0.185	0.140	0.015	0.003	0.002	0.000	0.000
2017	0.070	0.092	0.581	0.146	0.105	0.007	0.000	0.000	0.000
2018	0.086	0.102	0.071	0.581	0.098	0.053	0.009	0.000	0.001
2019	0.105	0.640	0.085	0.038	0.110	0.016	0.006	0.000	0.000
2020	0.286	0.274	0.304	0.048	0.014	0.058	0.010	0.003	0.003
2021	0.060	0.608	0.185	0.114	0.014	0.006	0.010	0.001	0.001
2022	0.087	0.292	0.412	0.100	0.085	0.010	0.004	0.007	0.003
2023	0.012	0.429	0.260	0.207	0.057	0.028	0.004	0.001	0.002
2024	0.012	0.239	0.441	0.162	0.109	0.031	0.004	0.001	0.001
2025	0.042	0.167	0.257	0.320	0.138	0.067	0.006	0.002	0.001

Table 13. Observed mean weight-at-age in grams (g) of Pacific Herring for nearshore samples from 2015 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

	Mean weight-at-age (g)								
Year	2	3	4	5	6	7	8	9	10
2015	51	67	76	81	109	104	NA	NA	NA
2016	49	66	72	81	88	83	88	NA	NA
2017	58	70	85	91	94	98	NA	NA	NA
2018	53	77	85	95	103	107	112	NA	114
2019	54	70	81	100	111	116	118	NA	NA
2020	56	69	85	94	114	125	110	131	120
2021	57	71	82	101	111	90	124	121	121
2022	57	71	89	100	118	126	123	128	148
2023	44	72	84	97	108	119	123	142	122
2024	47	69	85	98	103	114	105	110	125
2025	58	71	92	107	118	122	141	128	119

Table 14. Observed mean length-at-age in millimetres (mm) of Pacific Herring for nearshore samples from 2015 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

	Mean length-at-age (mm)								
Year	2	3	4	5	6	7	8	9	10
2015	154	170	178	182	199	200	NA	NA	NA
2016	154	171	178	186	188	197	192	NA	NA
2017	166	178	190	193	197	201	NA	NA	NA
2018	159	179	188	196	200	202	211	NA	220
2019	155	170	179	191	198	203	201	NA	NA
2020	156	169	180	186	200	205	197	214	204
2021	156	172	180	193	202	188	208	209	197
2022	165	176	189	196	205	209	208	217	222
2023	146	169	179	187	194	198	200	212	208
2024	150	170	183	191	196	201	198	203	219
2025	155	167	181	191	196	200	205	204	204

Table 15. Observed proportion-at-age for Pacific Herring from 2015 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older.

	Proportion-at-age								
Year	2	3	4	5	6	7	8	9	10
2015	0.140	0.238	0.505	0.068	0.017	0.028	0.004	0.000	0.001
2016	0.040	0.648	0.168	0.124	0.014	0.003	0.004	0.001	0.000
2017	0.025	0.076	0.664	0.132	0.082	0.016	0.002	0.002	0.000
2018	0.053	0.133	0.109	0.566	0.096	0.035	0.005	0.002	0.000
2019	0.055	0.504	0.097	0.049	0.232	0.040	0.018	0.003	0.000
2020	0.180	0.275	0.347	0.047	0.024	0.100	0.022	0.004	0.000
2021	0.173	0.533	0.136	0.118	0.014	0.009	0.013	0.003	0.002
2022	0.073	0.292	0.436	0.089	0.097	0.005	0.001	0.005	0.001
2023	0.065	0.501	0.222	0.146	0.039	0.019	0.004	0.004	0.000
2024	0.006	0.221	0.487	0.160	0.100	0.022	0.003	0.001	0.001
2025	0.033	0.183	0.223	0.345	0.125	0.075	0.015	0.001	0.000

Table 16. Pacific Herring spawn survey locations, start date, and spawn index in metric tonnes (t) in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (NAs).

Statistical Area	Section	Location name	Start date	Spawn index (t)
23	232	David Is	March 09	470
23	232	Equis Beach	March 11	744
23	232	Erin Isl	March 10	291
23	232	Forbes Is	March 09	649
23	232	Hermit Isl	March 09	219
23	232	Larkin Is	March 09	2,140
23	232	Lyall Pt	March 10	761
23	232	Macoah Pass	February 10	2,590
23	232	Maggie Rvr	March 07	751
23	232	Mayne Bay	March 10	472
23	232	Nettle Is	March 10	2,956
23	232	Newcombe Chnl	March 09	851
23	232	Ottaway Islet	March 09	402
23	232	Pinkerton Is	March 10	62
23	232	Spilling Islet	March 09	390
23	232	Spring Cv	March 10	106
23	232	St Ines Is	March 09	1,328
23	232	Stopper Is	March 09	3,734
23	232	Stuart Bay	March 09	82
23	232	Toquart Bay	March 09	334
23	232	Two Rivers	February 10	5,963
23	232	Ucluelet	April 08	NA
24	242	Antons Spit	February 13	628
24	242	Hesquiat Hrbr	January 09	666
24	242	Hesquiat Hrbr Hd	March 18	247
24	242	Hesquiat Pen	March 13	720
24	242	Hesquiat Pt	March 09	185
24	242	Leclaire Pt	March 18	388
24	242	Rondeault Pt	March 09	91
24	243	George Is	March 03	542
24	243	Hot Springs Cv	March 14	975
24	244	Bawden Bay	February 28	NA
24	244	Flores Is E	March 17	316
24	244	Matilda Inlt	March 17	3,492
24	244	McNeil Pen	March 12	1,455
24	245	Bartlett Isl	March 24	NA
24	245	Chetarpe Reserve	March 13	957
24	245	Clifford Pt	March 18	298

Table 16 continued

Statistical Area	Section	Location name	Start date	Spawn index (t)
24	245	Whitesand Cv	March 11	3,328
25	251	Bajo Pt	March 06	NA
25	251	Skuna Bay	March 09	NA
25	252	Boca del Infierno Bay	March 03	118
25	252	Cook Chnl	March 03	602
25	252	Friendly Cv	February 28	195
25	252	Marvinas Bay & Is	March 06	510
25	252	McKay Psg	March 03	586
25	252	Pantoja Is	March 06	429
25	252	Saavedra Is	March 03	1,899
25	252	Santa Gertrudis Cv	March 03	579
25	253	Nuchatlitz Inner	March 03	45
25	253	Nuchatlitz Outer	March 03	218
25	253	Port Langford	February 06	2,630
25	253	Rosa Hrbr	March 28	96
25	253	Rosa Is	March 28	12

Table 17. Summary of Pacific Herring spawn survey data from 2015 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Units: metres (m), and metric tonnes (t).

Year	Total length (m)	Mean width (m)	Mean number of egg layers	Spawn index (t)
2015	20,450	275	1.1	11,323
2016	60,575	122	1.4	20,527
2017	44,200	180	1.3	16,476
2018	61,825	136	1.5	28,107
2019	49,325	130	1.2	17,029
2020	44,125	162	0.9	18,760
2021	45,851	203	1.1	29,338
2022	64,925	168	1.3	23,706
2023	74,850	195	1.9	77,005
2024	102,550	160	2.5	86,307
2025	117,950	134	1.6	47,525

8 Figures

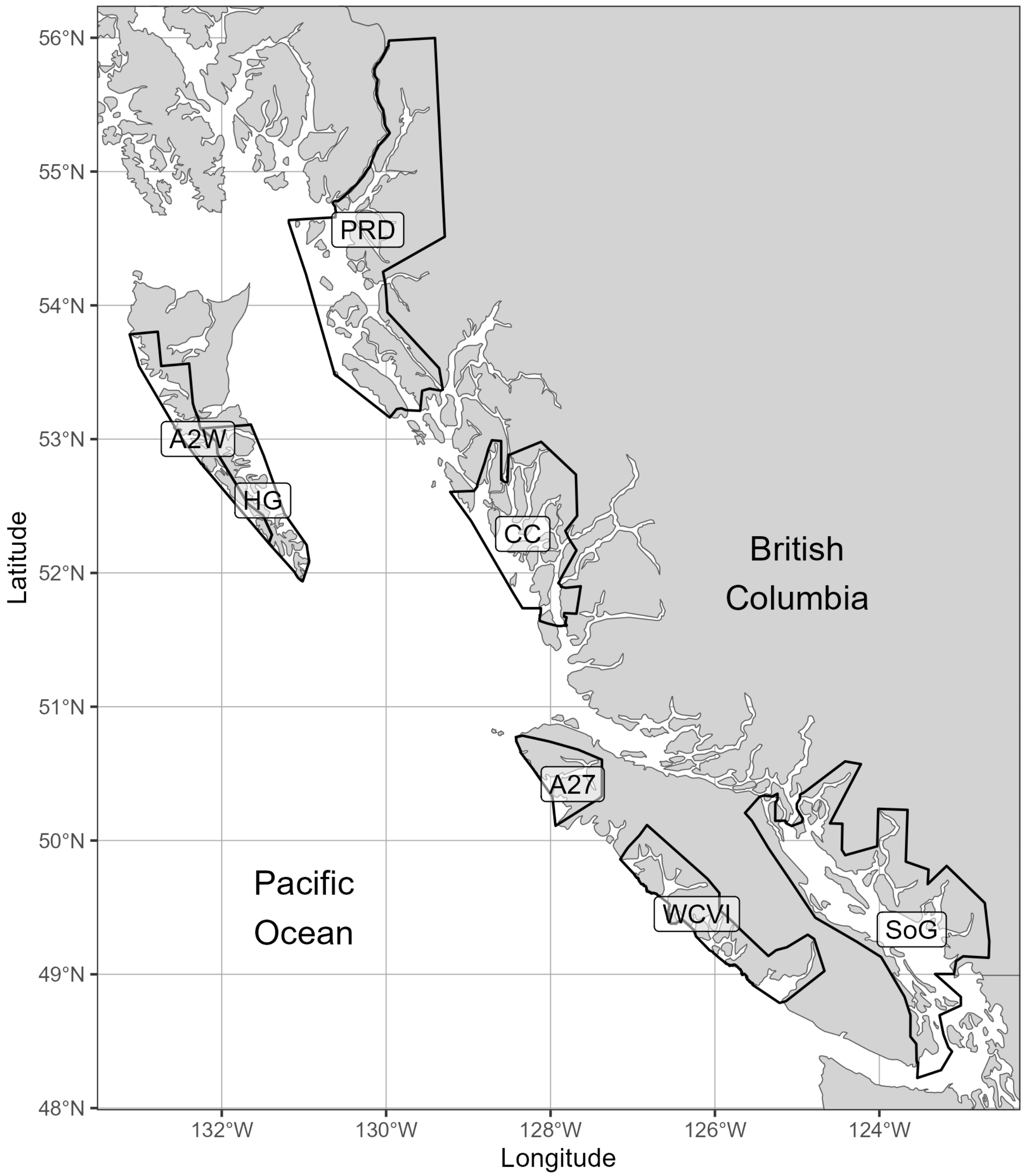


Figure 1. Boundaries for the Pacific Herring stock assessment regions (SARs) in British Columbia. There are 5 major SARs: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), Strait of Georgia (SoG), and West Coast of Vancouver Island (WCVI). There are 2 minor SARs: Area 27 (A27) and Area 2 West (A2W). Units: kilometres (km).

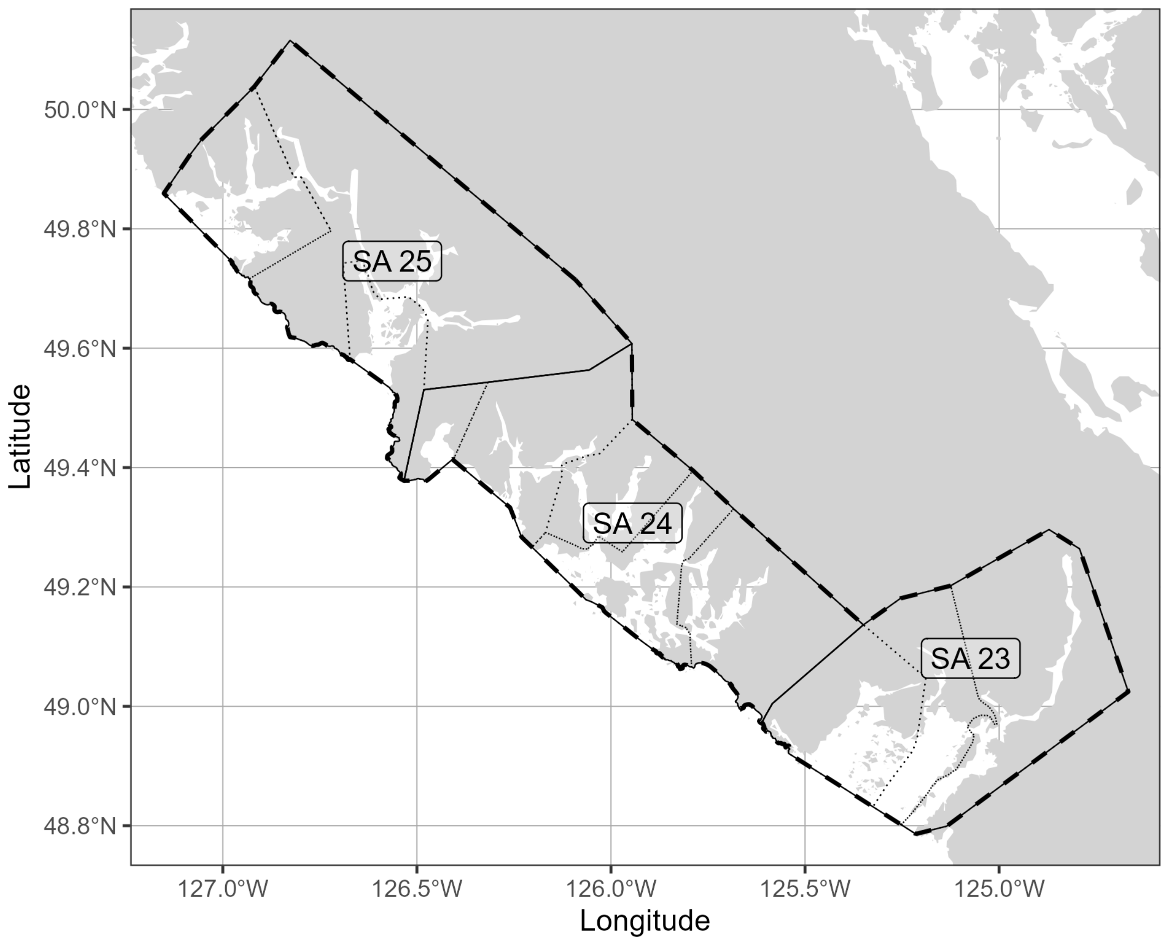


Figure 2. Boundaries for the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), associated Statistical Areas (SA; thin solid lines), and associated Sections (thin dotted lines). Units: kilometres (km).

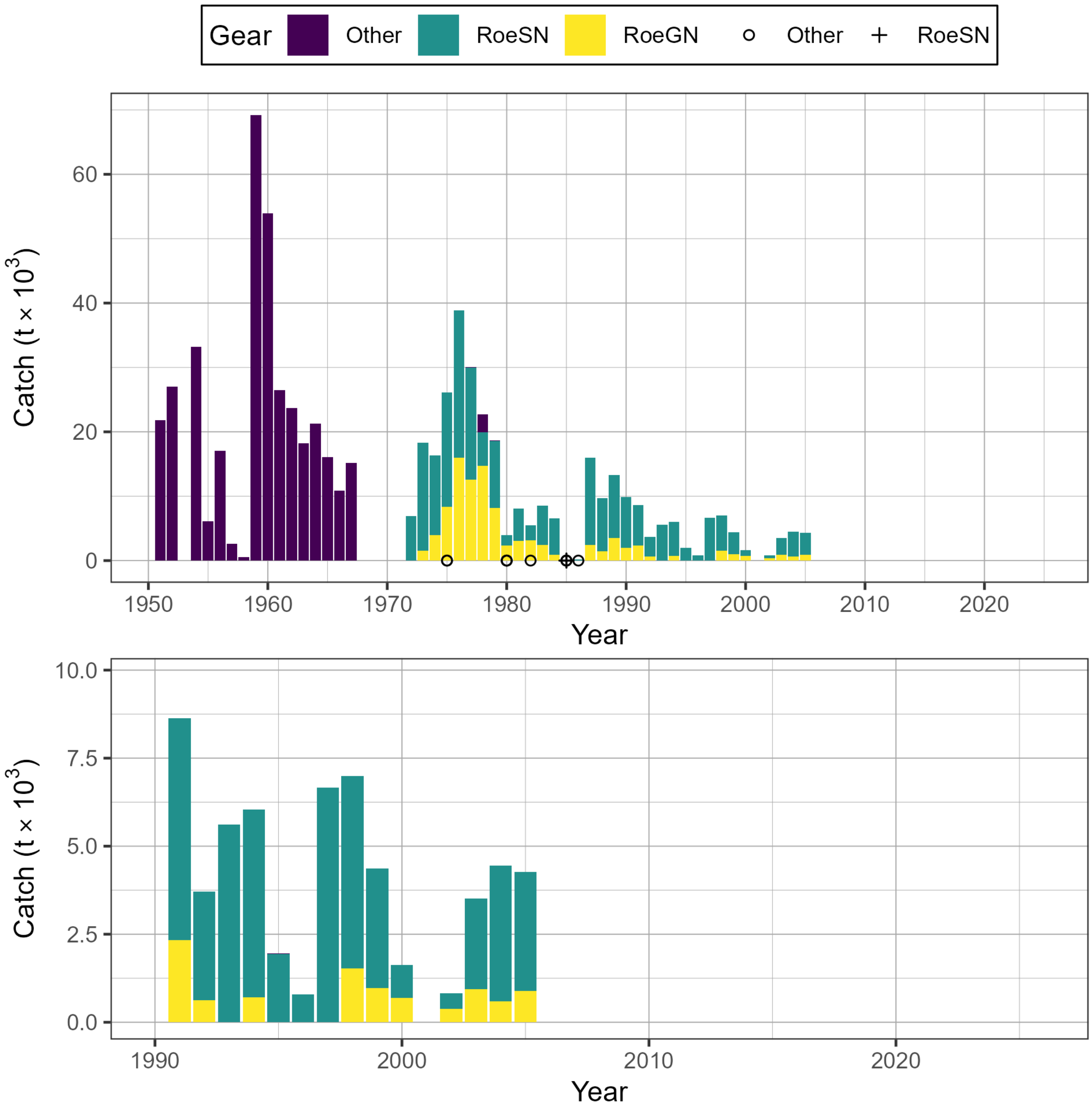


Figure 3. Time series of total landed catch in thousands of metric tonnes (t $\times$ 10^3) of Pacific Herring by gear type from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). Legend: ‘Other’ represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; ‘RoeSN’ represents the roe seine fishery; and ‘RoeGN’ represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Bottom panel shows catch since 1990 in more detail. Note: symbols indicate years in which catch by gear type (i.e., Other, RoeSN, RoeGN) is withheld due to privacy concerns.

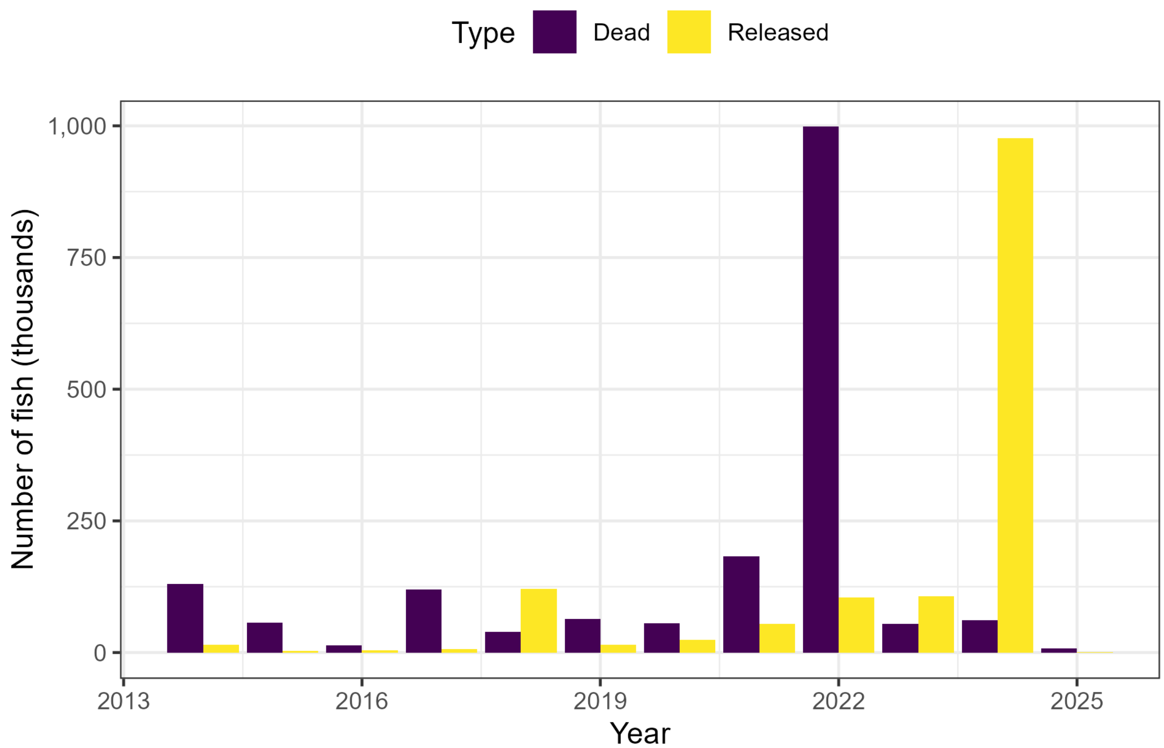


Figure 4. Incidental Pacific Herring mortality in aquaculture activities in thousands of fish from 2014 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). Note: figure may include data outside SAR boundaries.

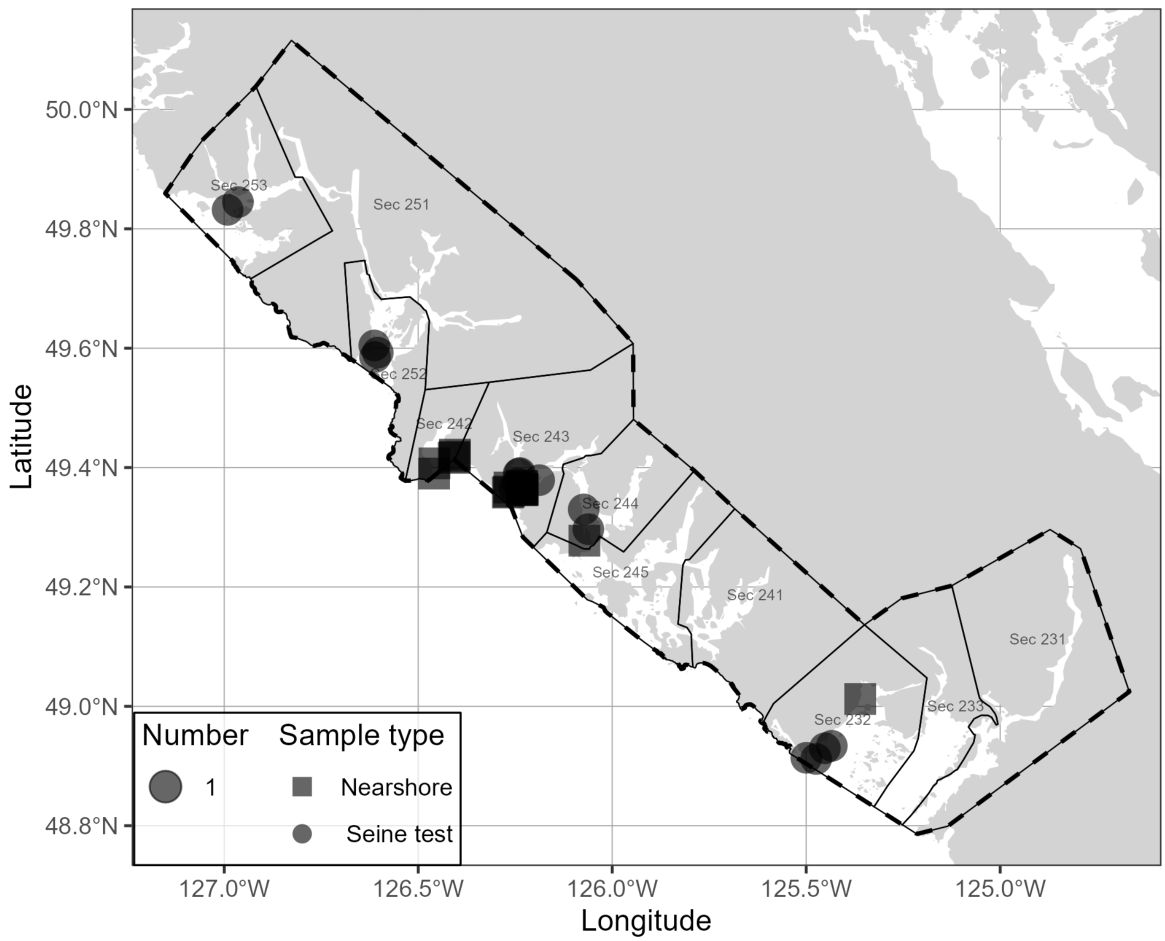


Figure 5. Location and type of Pacific Herring biological samples collected in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). Units: kilometres (km).

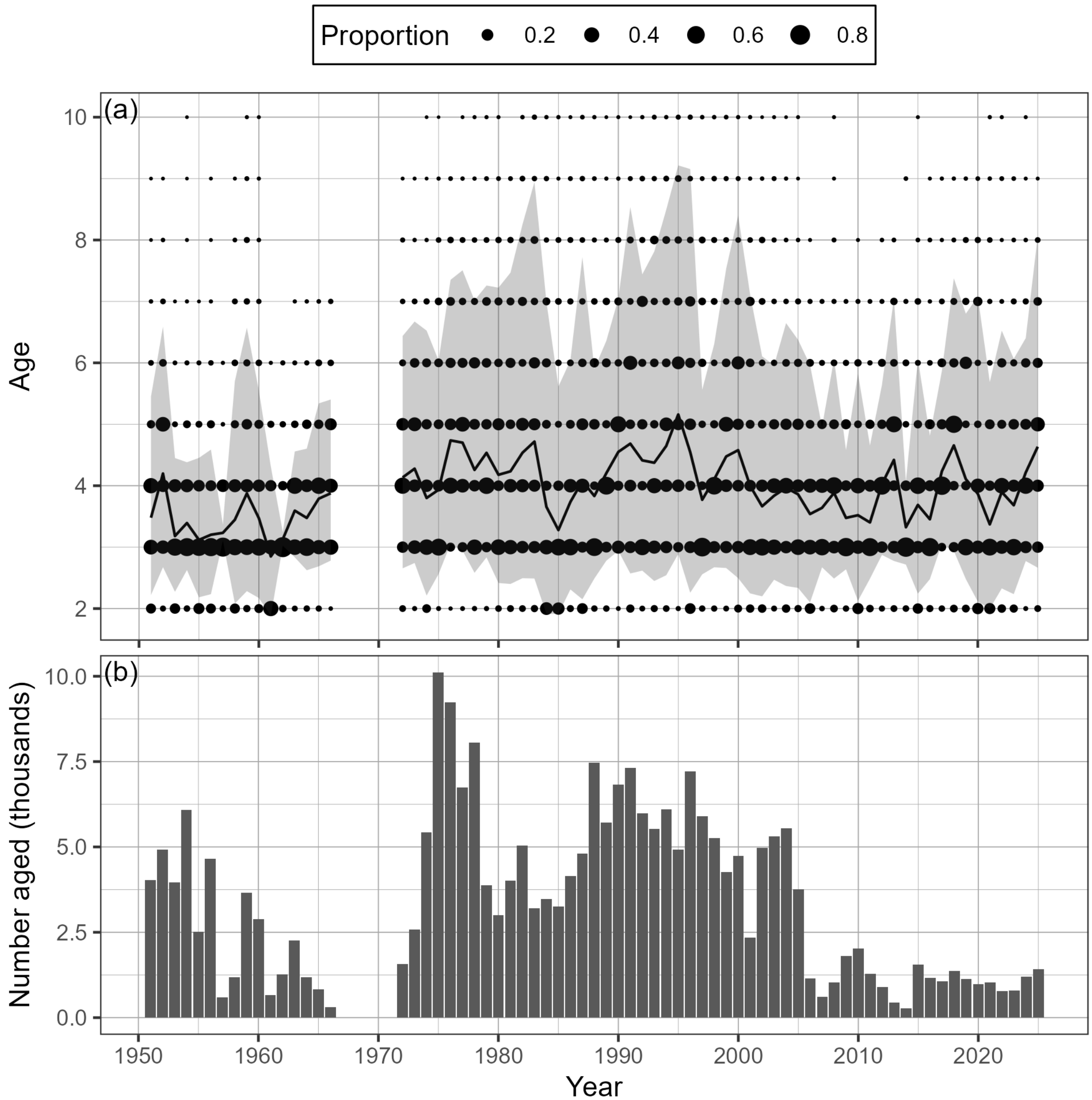


Figure 6. Time series of observed proportion-at-age (a) and number aged in thousands (b) of Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The black line is the mean age, and the shaded area is the approximate 90% distribution. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a 'plus group' which includes fish ages 10 and older.

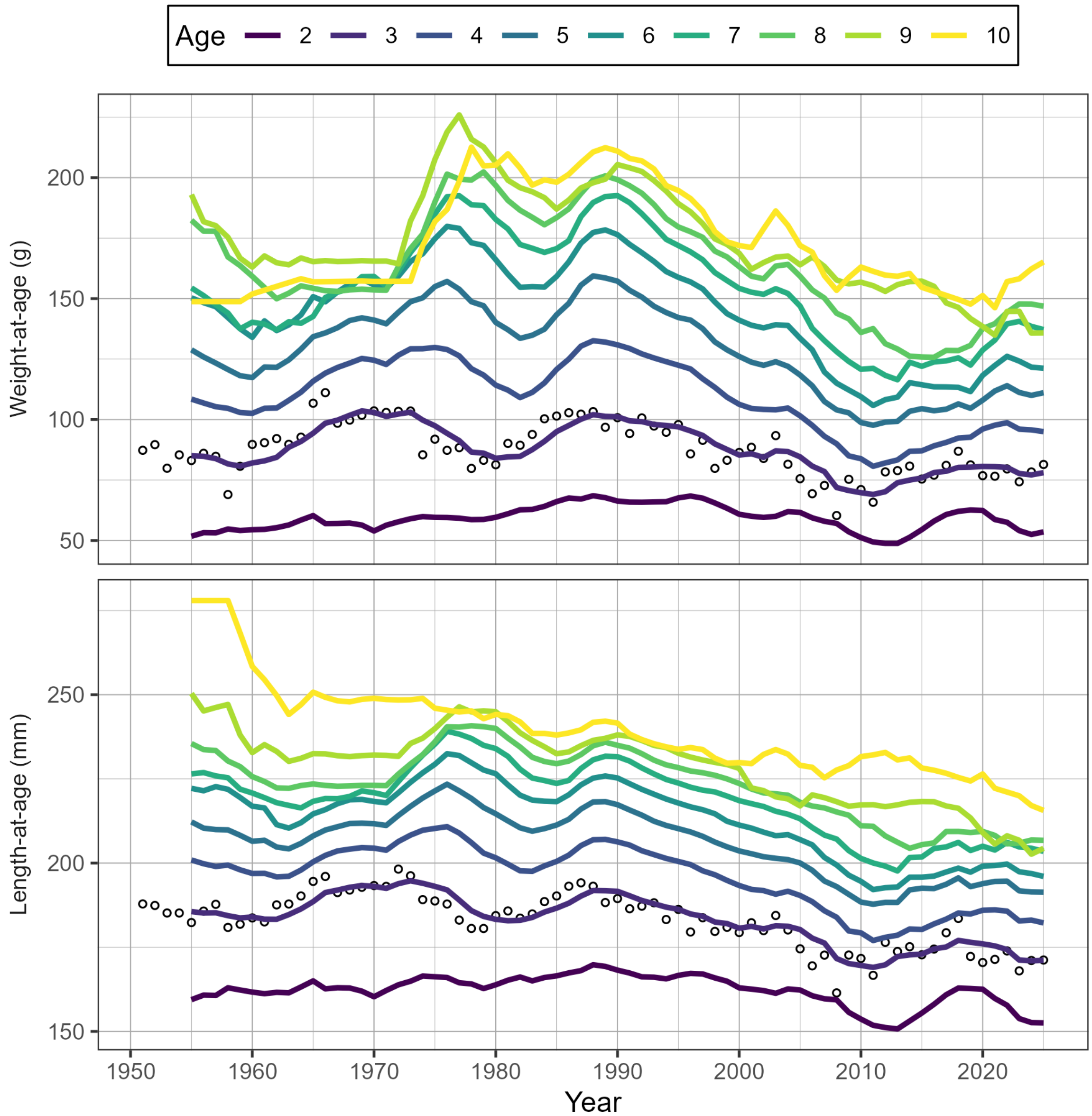


Figure 7. Time series of weight-at-age in grams (g) and length-at-age in milimetres (mm) for age-3 (circles) and 5-year running mean weight- and length-at-age (lines) for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). Missing weight- and length-at-age values (i.e., years with no biological samples) are imputed using one of two methods: missing values at the beginning of the time series are imputed by extending the first non-missing value backwards; other missing values are imputed as the mean of the previous 5 years. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a 'plus group' which includes fish ages 10 and older.

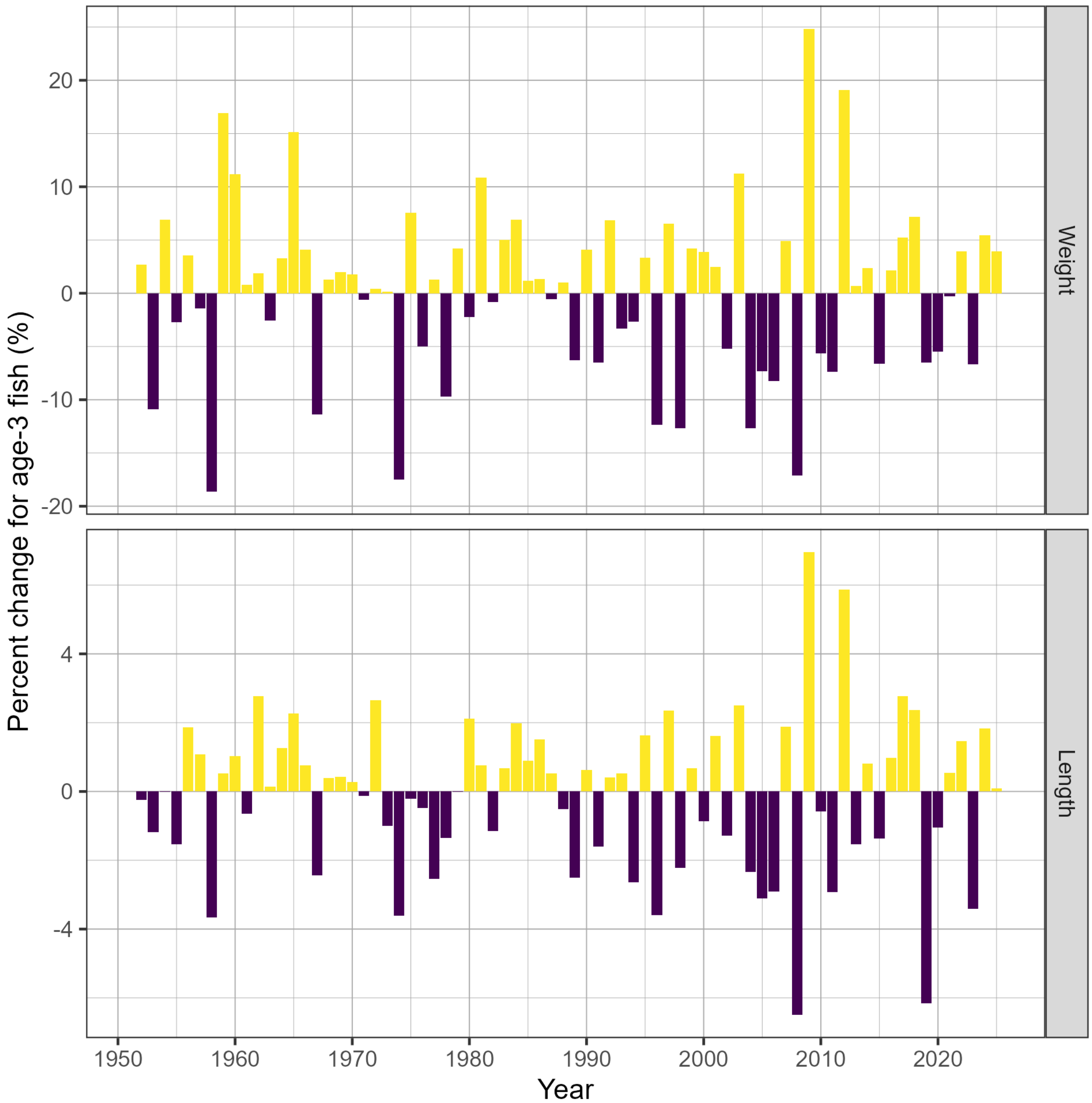


Figure 8. Time series of percent change (%) in weight and length for age-3 fish for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-3 fish, respectively, in year t. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

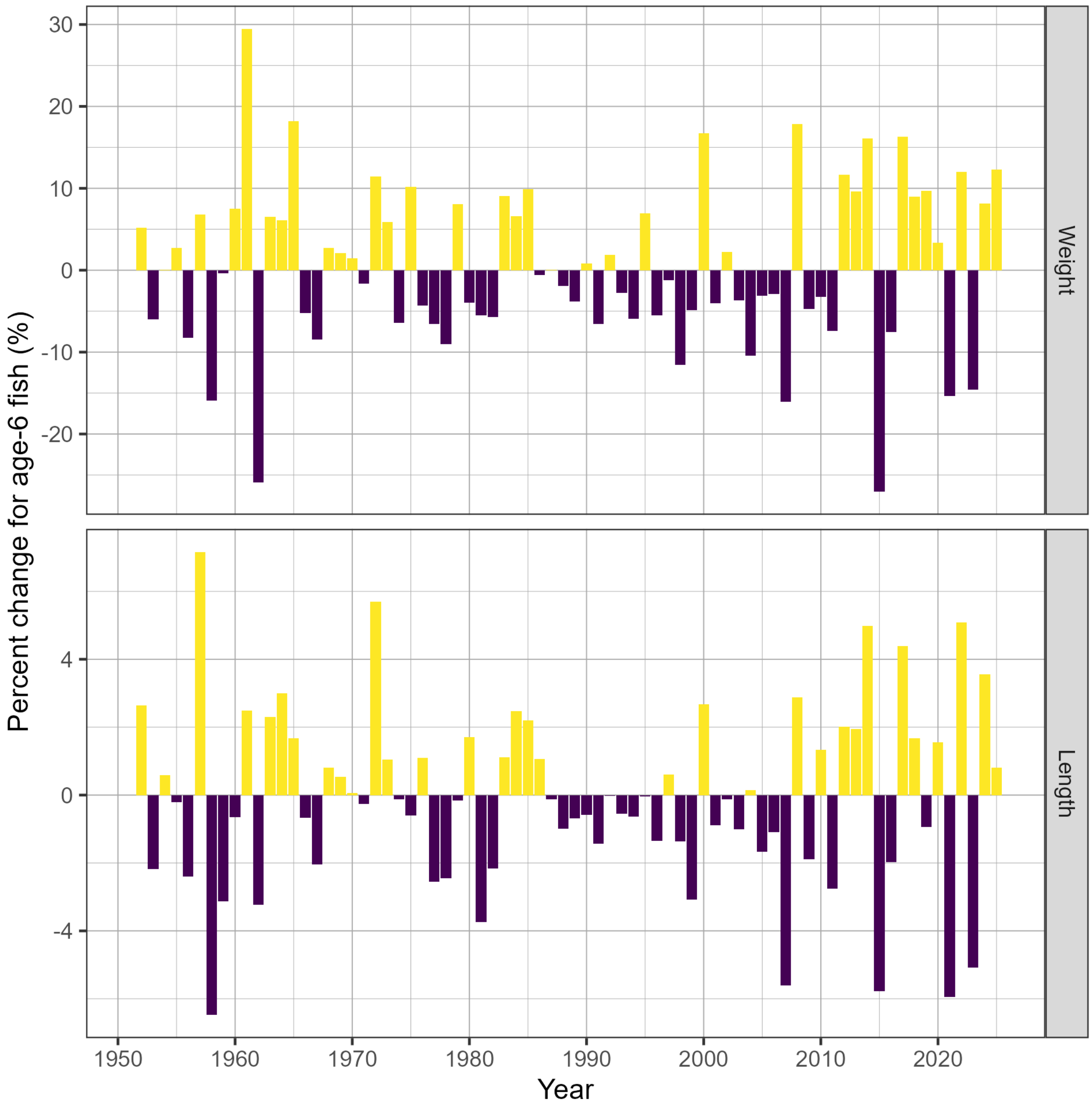


Figure 9. Time series of percent change (%) in weight and length for age-6 fish for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-6 fish, respectively, in year t. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

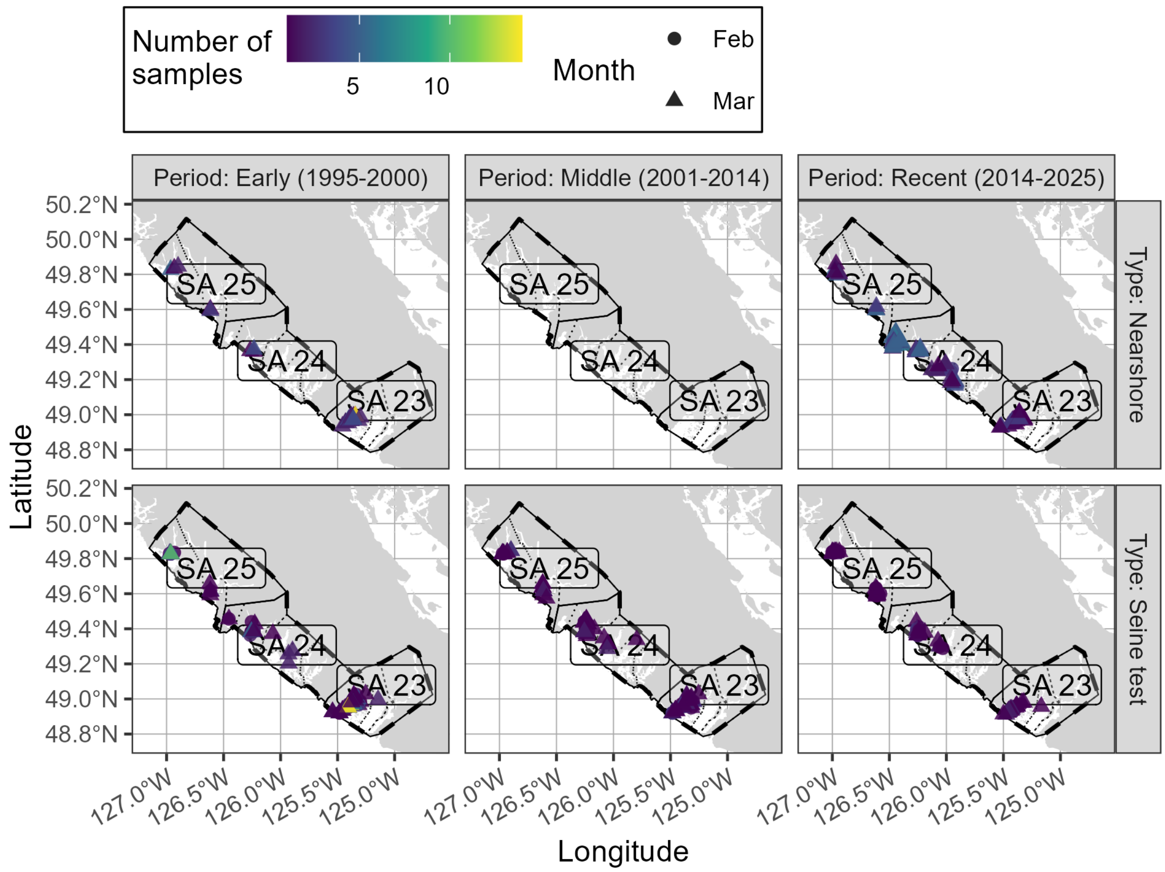


Figure 10. Location of Pacific Herring biological samples in the West Coast of Vancouver Island major stock assessment region (SAR) by period, sample type, and month. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

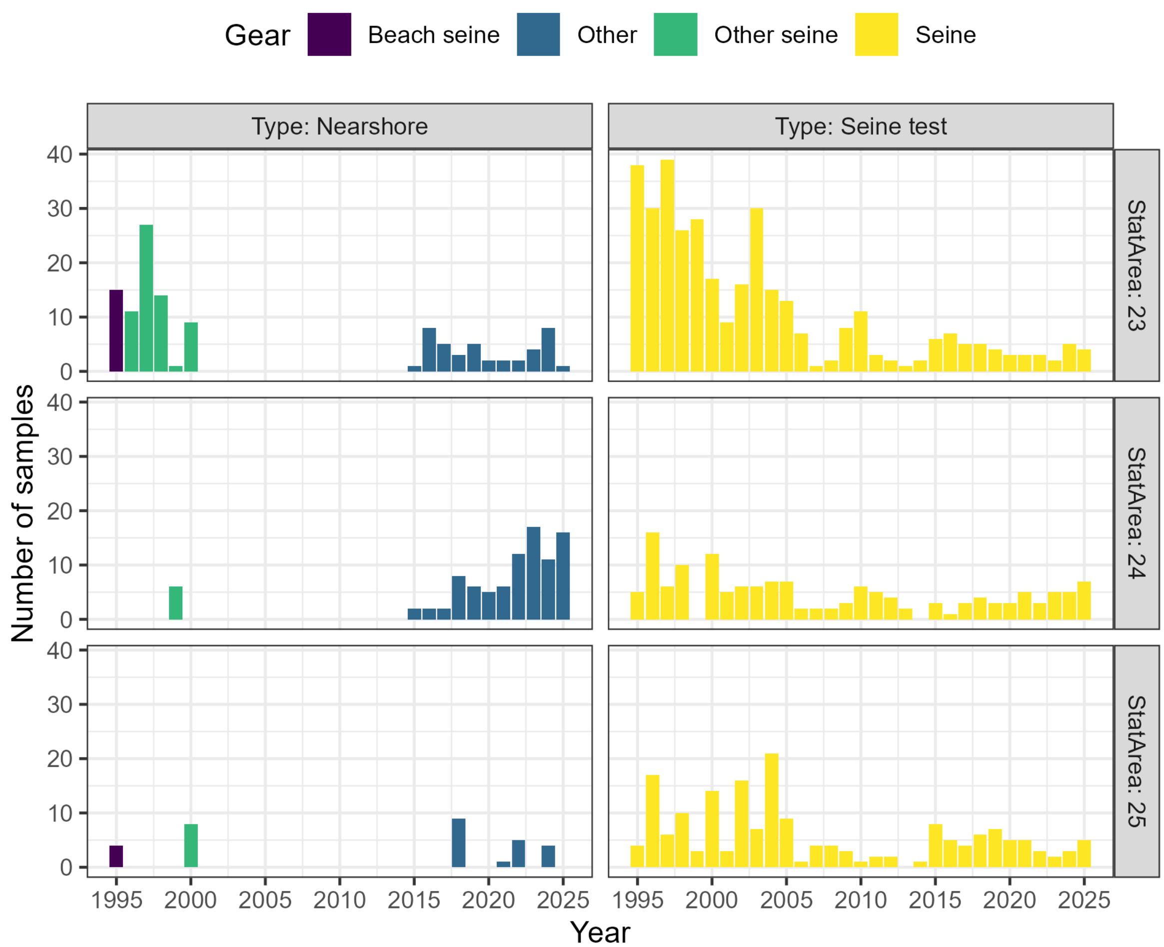


Figure 11. Number of Pacific Herring samples in the West Coast of Vancouver Island major stock assessment region (SAR) by sample type, statistical area, and gear type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

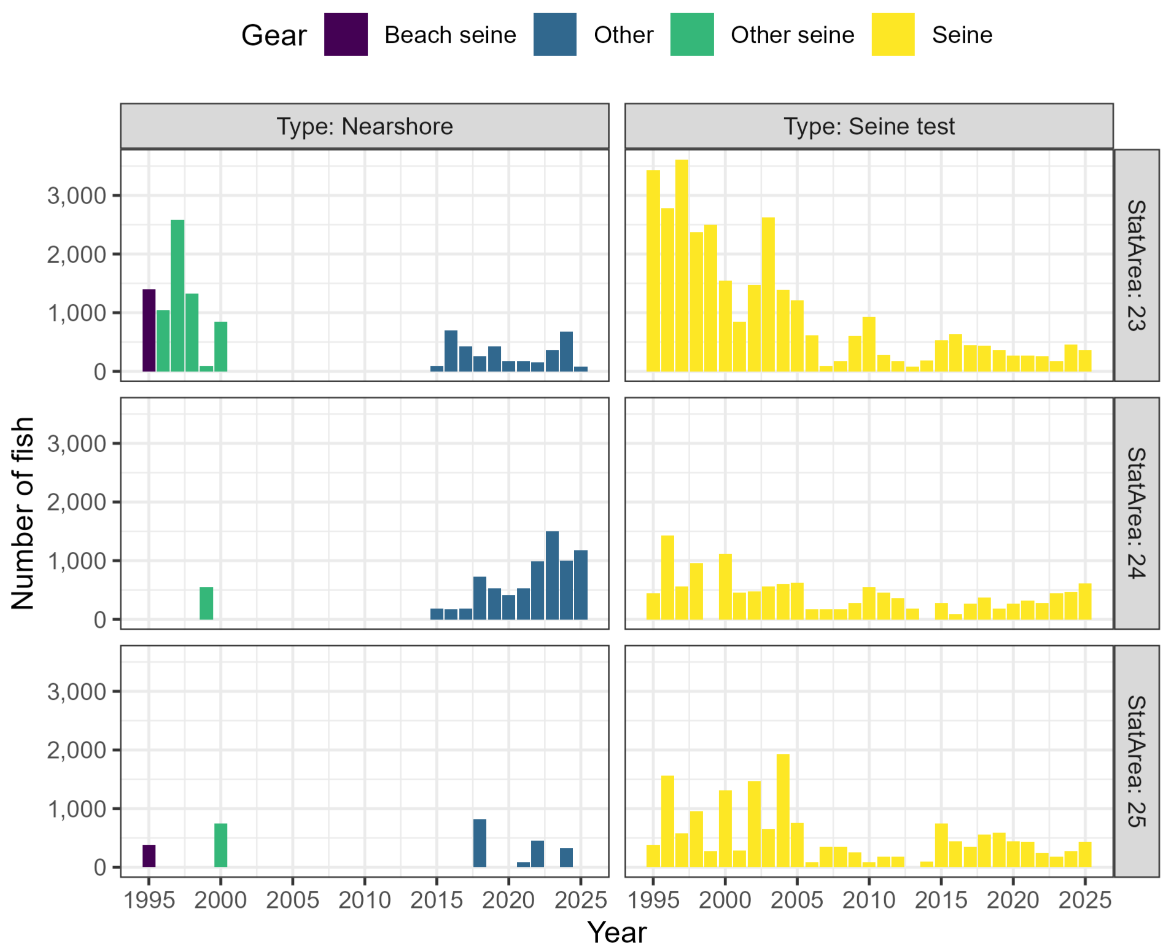


Figure 12. Number of Pacific Herring in biological samples in the West Coast of Vancouver Island major stock assessment region (SAR) by sample type, statistical area, and gear type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

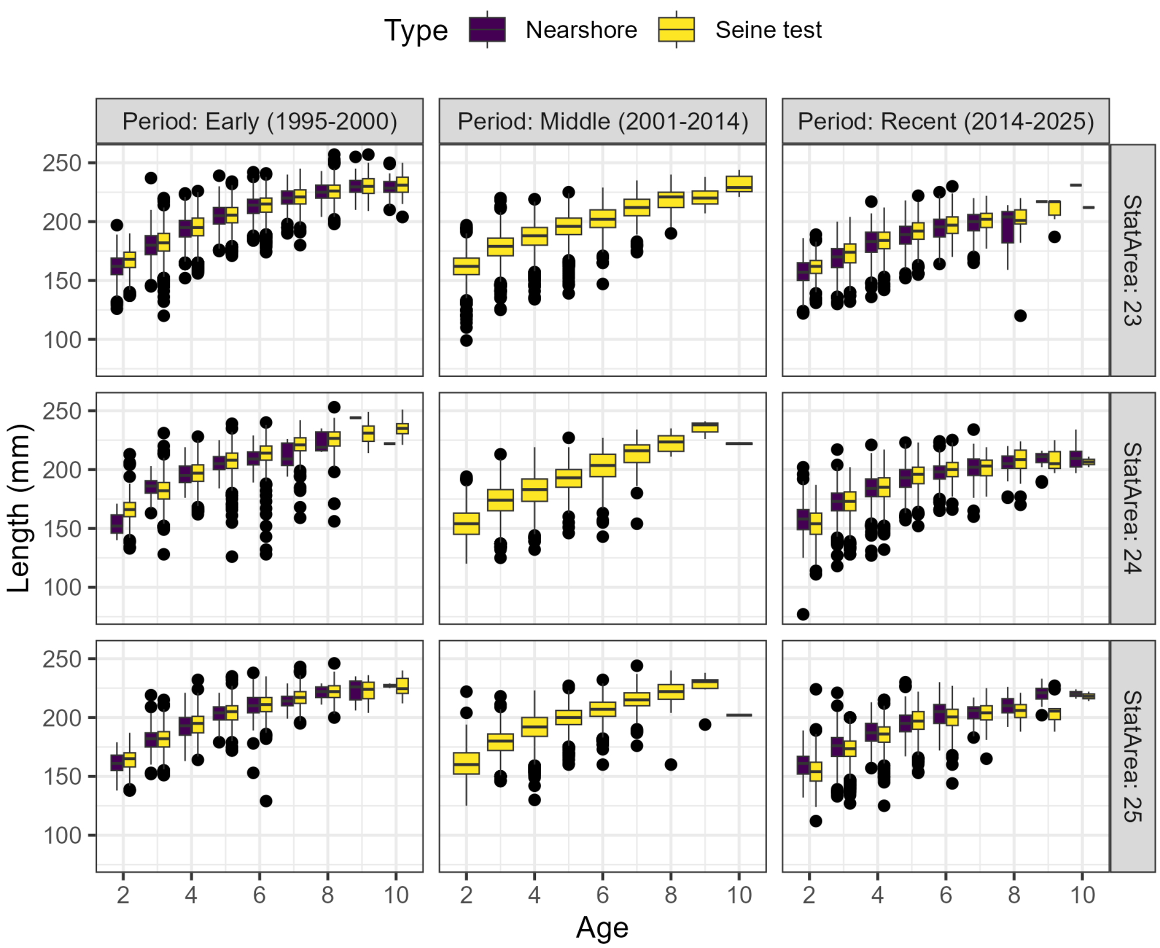


Figure 13. Length-at-age in millimetres (mm) of Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR) by period, statistical area, and sample type. The outer edges of the boxes indicate the 25$^{\text{th}}$ and 75$^{\text{th}}$ percentiles, and the middle lines indicate the 50$^{\text{th}}$ percentiles (i.e., medians). The whiskers extend to $1.5 \times$ IQR, where IQR is the distance between the 25$^{\text{th}}$ and 75$^{\text{th}}$ percentiles, and dots indicate outliers. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a 'plus group' which includes fish ages 10 and older.

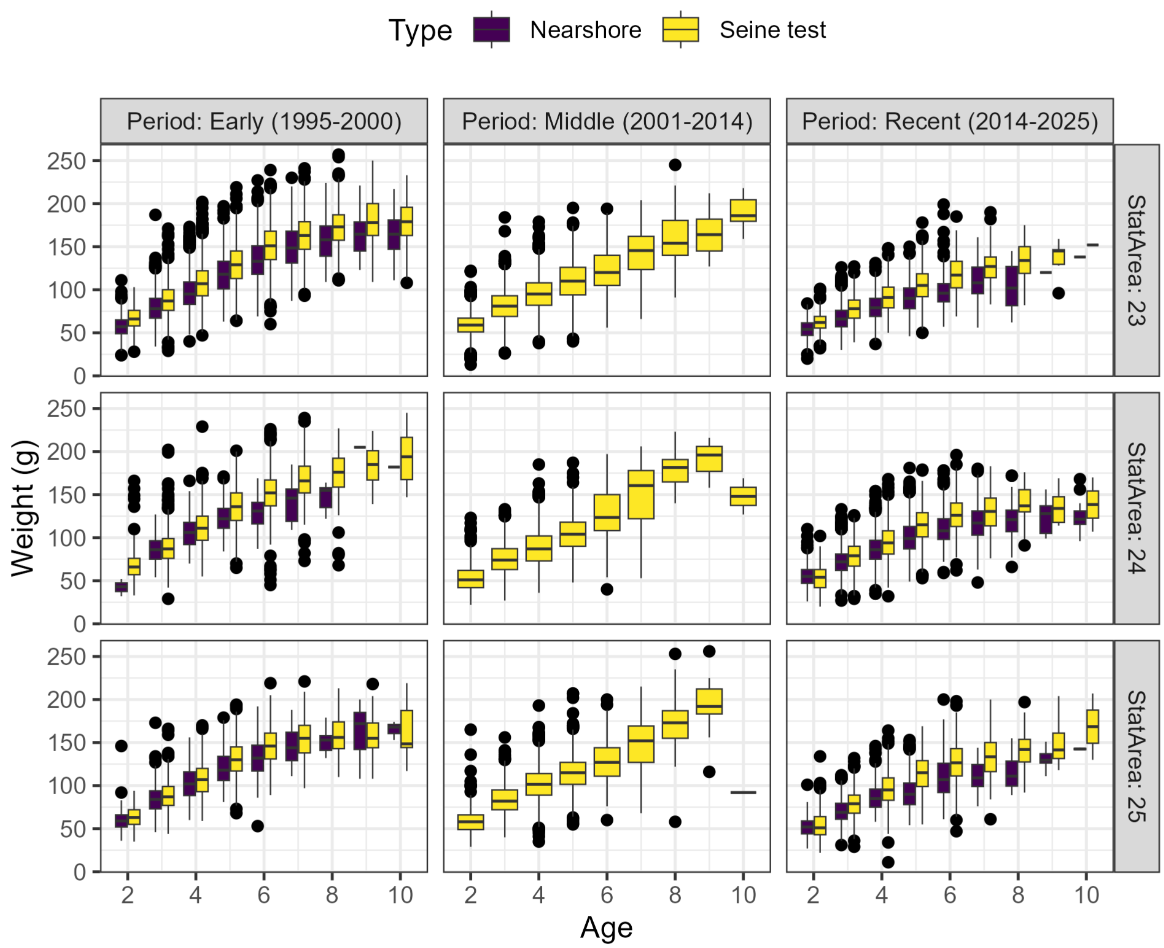


Figure 14. Weight-at-age in grams (g) of Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR) by period, statistical area, and sample type. The outer edges of the boxes indicate the 25$^{\mathrm{th}}$ and 75$^{\mathrm{th}}$ percentiles, and the middle lines indicate the 50$^{\mathrm{th}}$ percentiles (i.e., medians). The whiskers extend to 1.5 $\times$ IQR, where IQR is the distance between the 25$^{\mathrm{th}}$ and 75$^{\mathrm{th}}$ percentiles, and dots indicate outliers. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

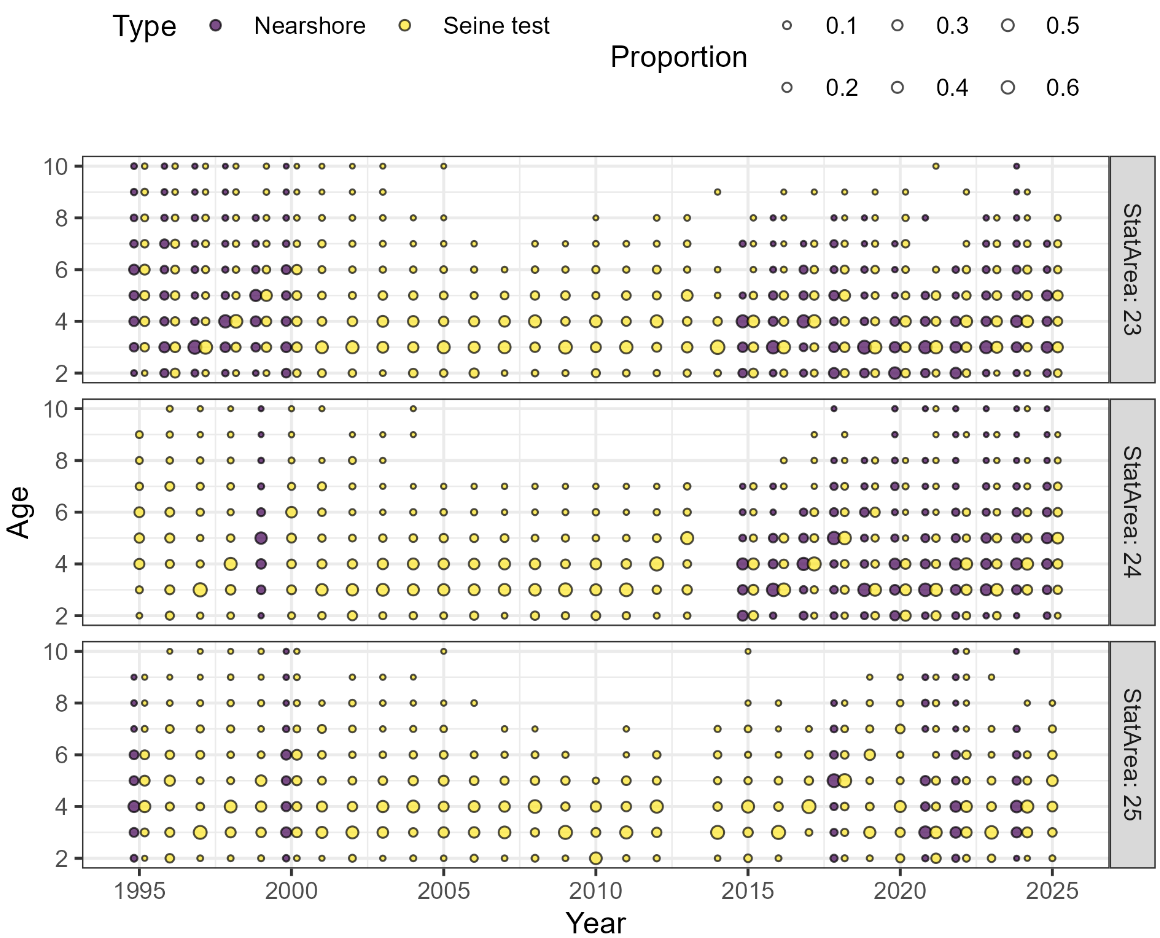


Figure 15. Proportion-at-age of Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR) by statistical area and sample type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a 'plus group' which includes fish ages 10 and older.

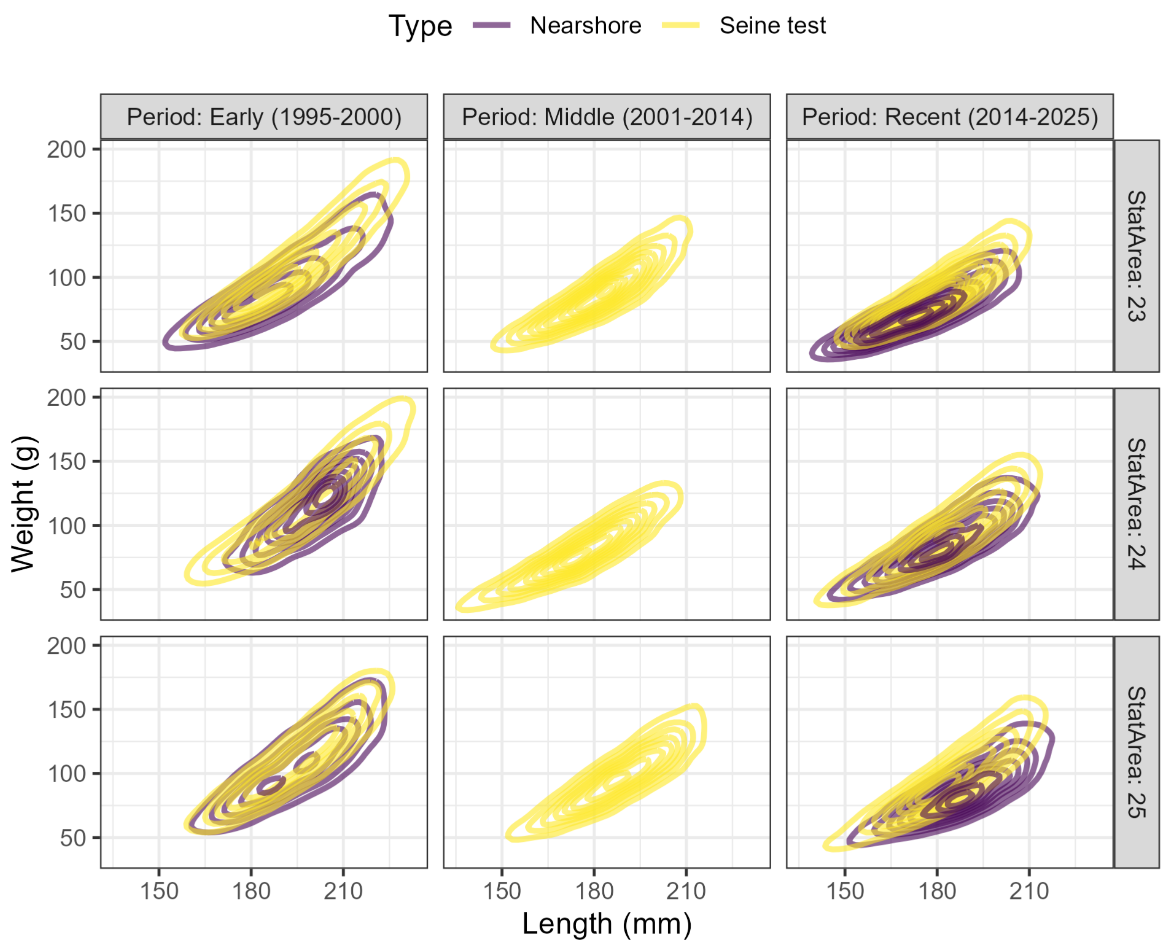


Figure 16. Length-weight relationship for Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR) by period, statistical area, and sample type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

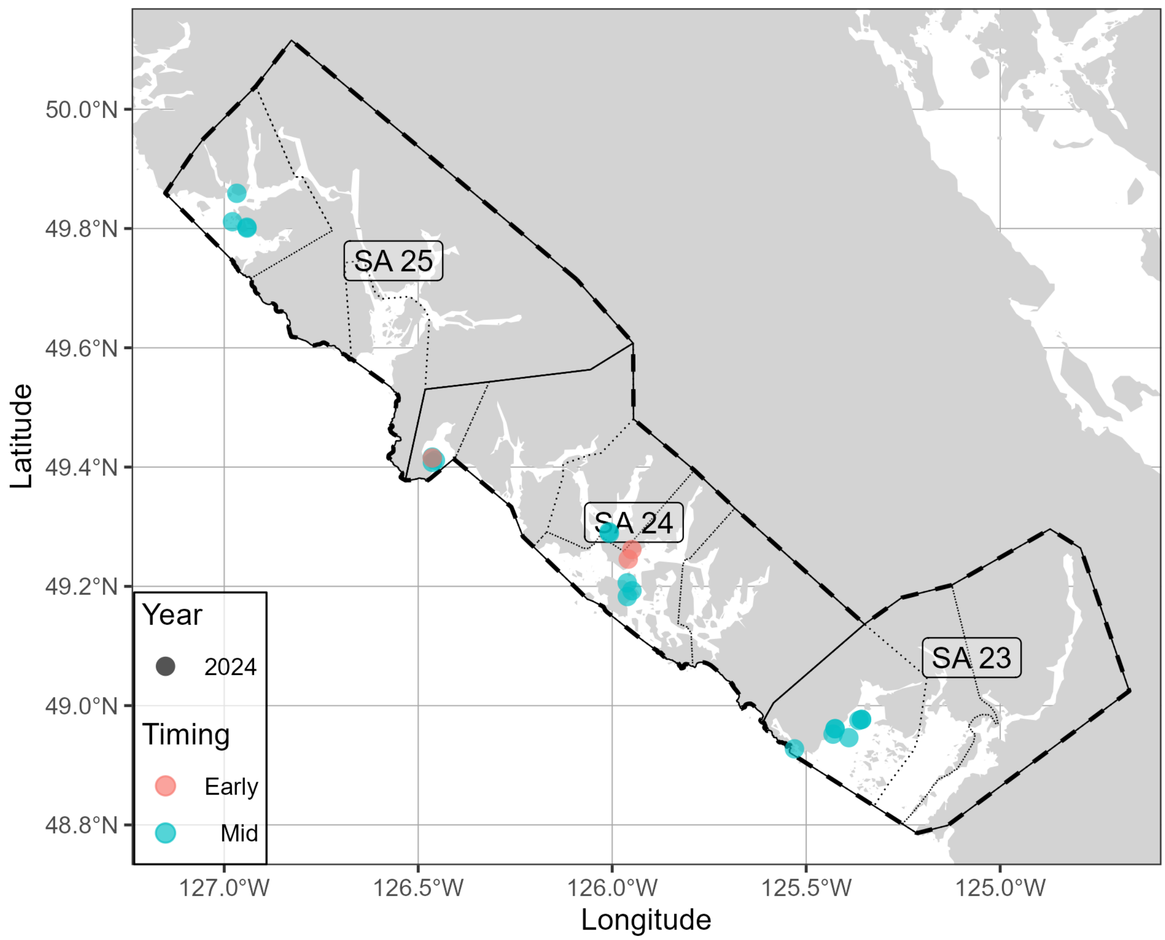


Figure 17. Nearshore genetics sampling in the West Coast of Vancouver Island major SAR. Note that timing indicates the collection date: 'Early' is January and February, 'Mid' is March and April, and 'Late' is May, June, and July. In addition, note that genetic samples from 2025 are not yet available. See Figure 2 for legend.

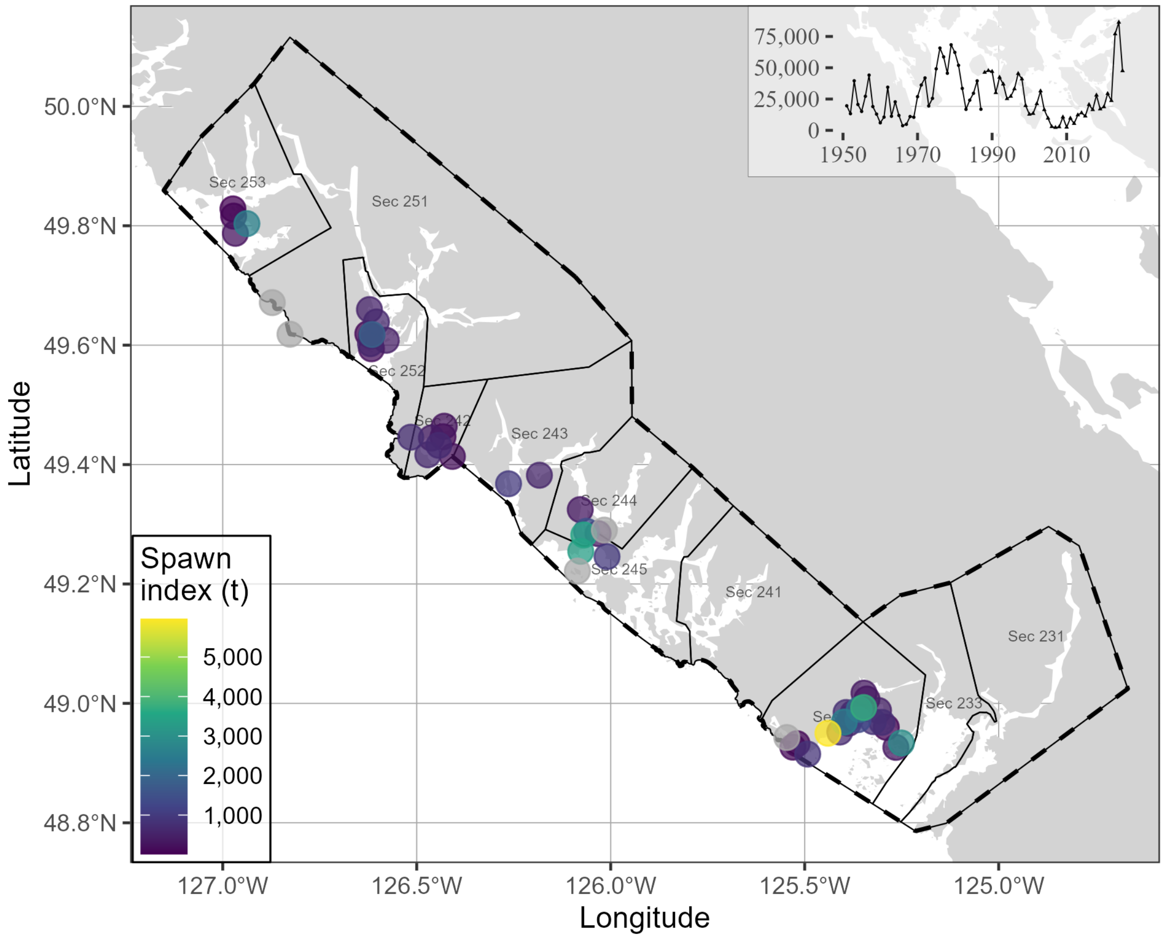


Figure 18. Pacific Herring spawn survey locations, and spawn index in metric tonnes (t) in 2025 in the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km).

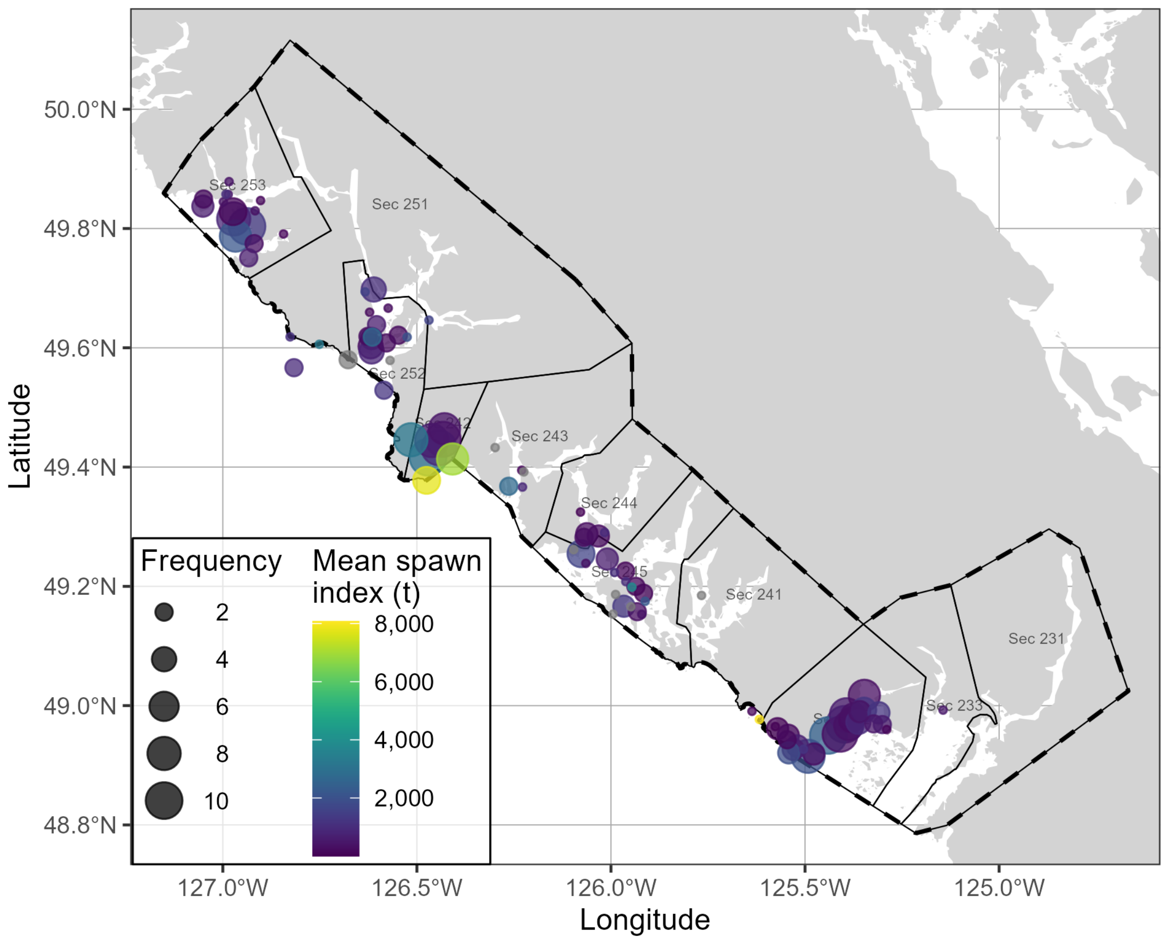


Figure 19. Pacific Herring spawn survey locations, mean spawn index in metric tonnes (t), and spawn frequency from 2015 to 2024 in the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Units: kilometres (km).

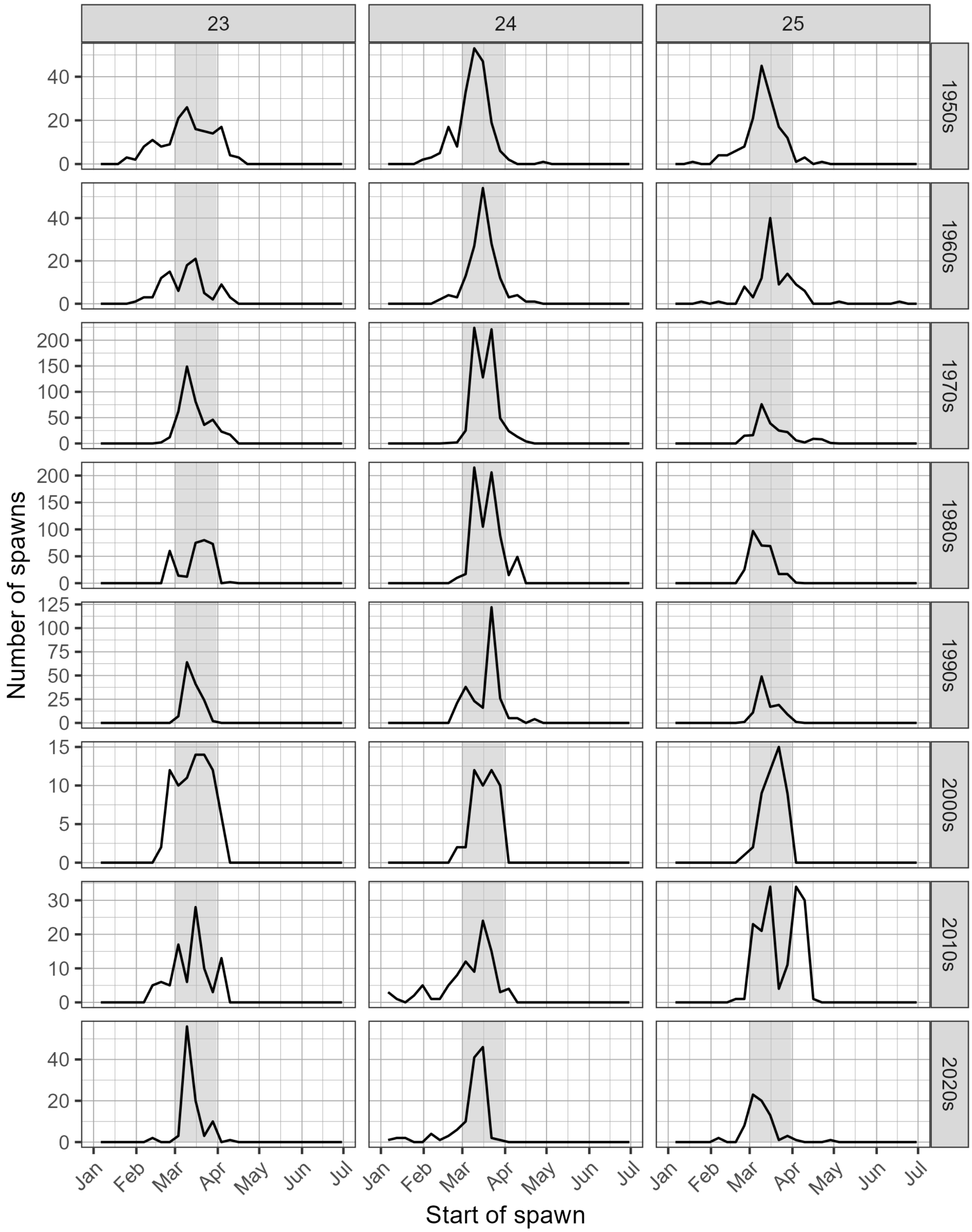


Figure 20. Pacific Herring spawn start date by decade and Statistical Area. Grey shaded regions indicate March 1st to 31st. Note that spawn size and intensity varies; therefore the number of spawns is not directly proportional to spawn extent or biomass.

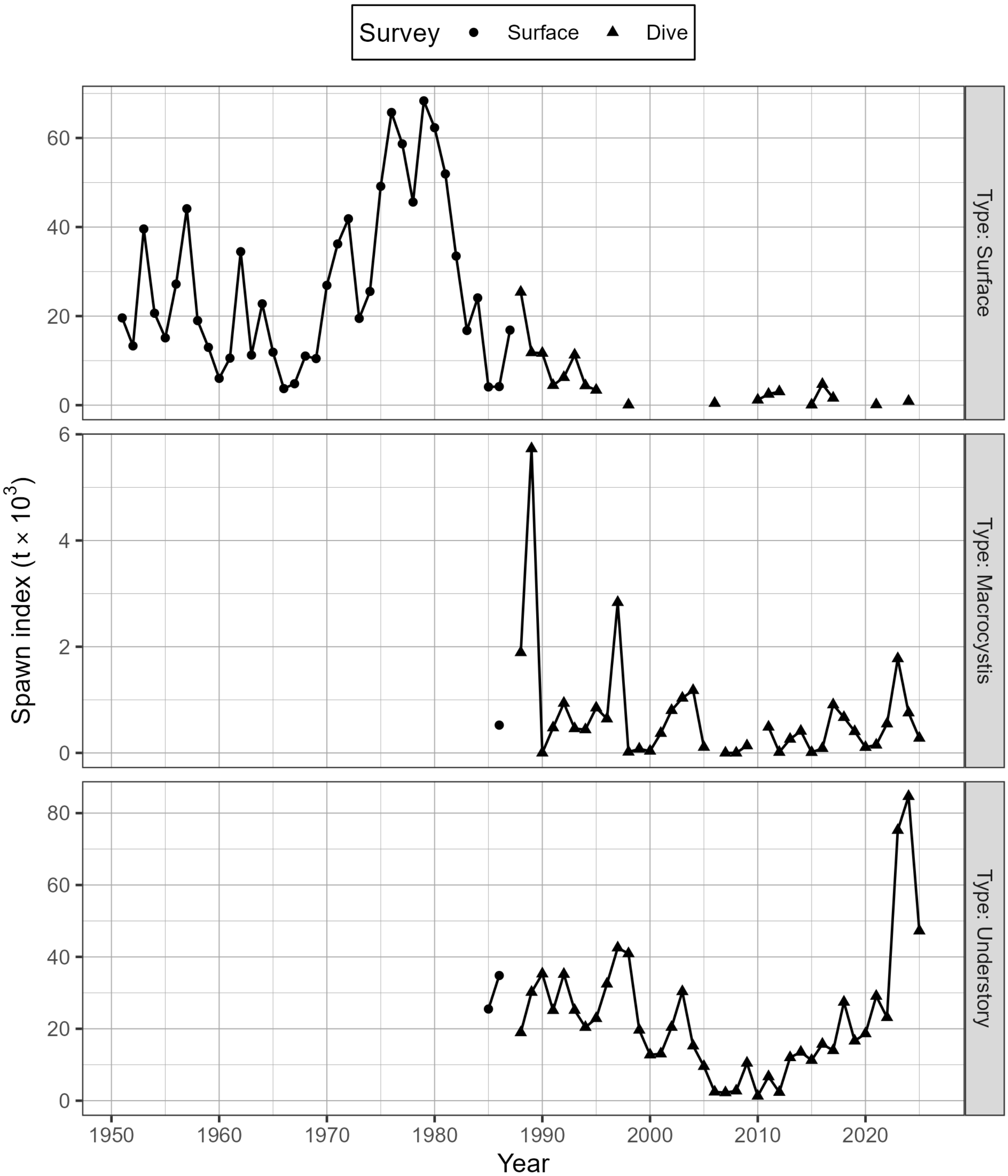


Figure 21. Time series of spawn index in thousands of metric tonnes (t $\times$ 10^3) by type for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). There are three types of spawn survey observations: observations of spawn taken from the surface usually at low tide, underwater observations of spawn on giant kelp, Macrocystis (*Macrocystis* spp.), and underwater observations of spawn on other types of algae and the substrate, which we refer to as 'understory.' The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

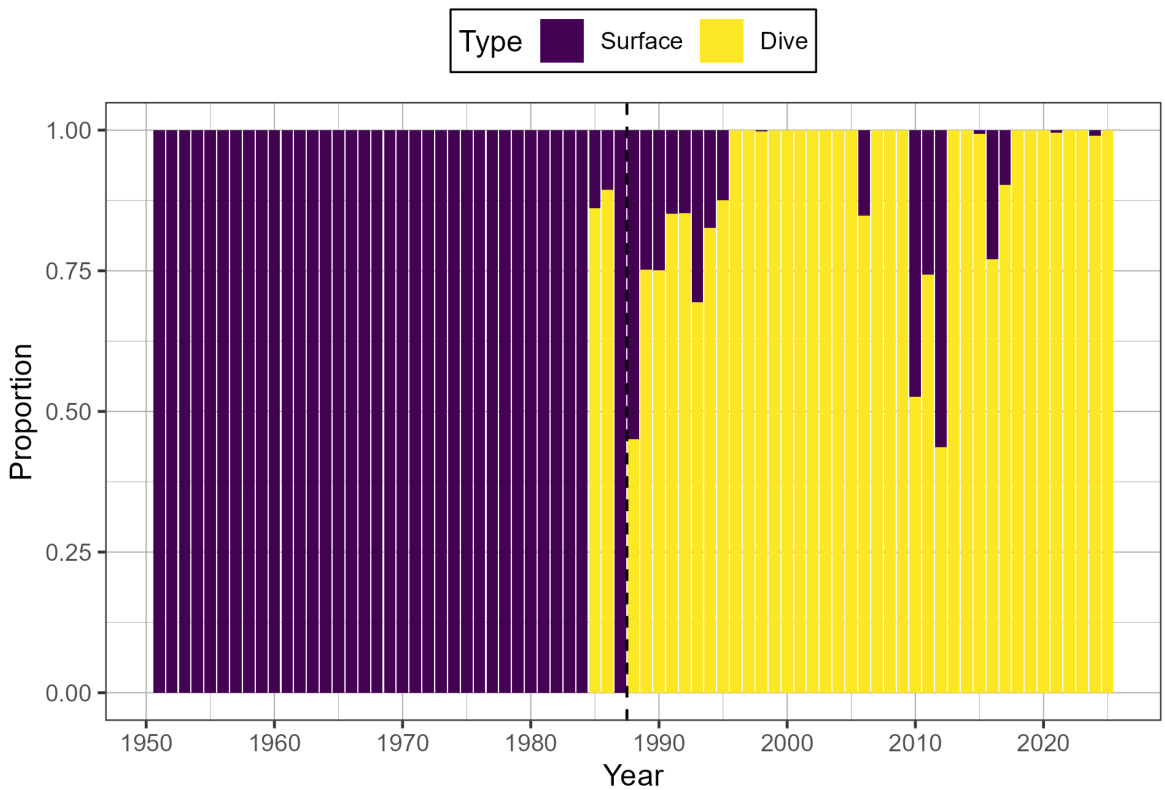


Figure 22. Time series of proportion of spawn index by surface and dive surveys for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

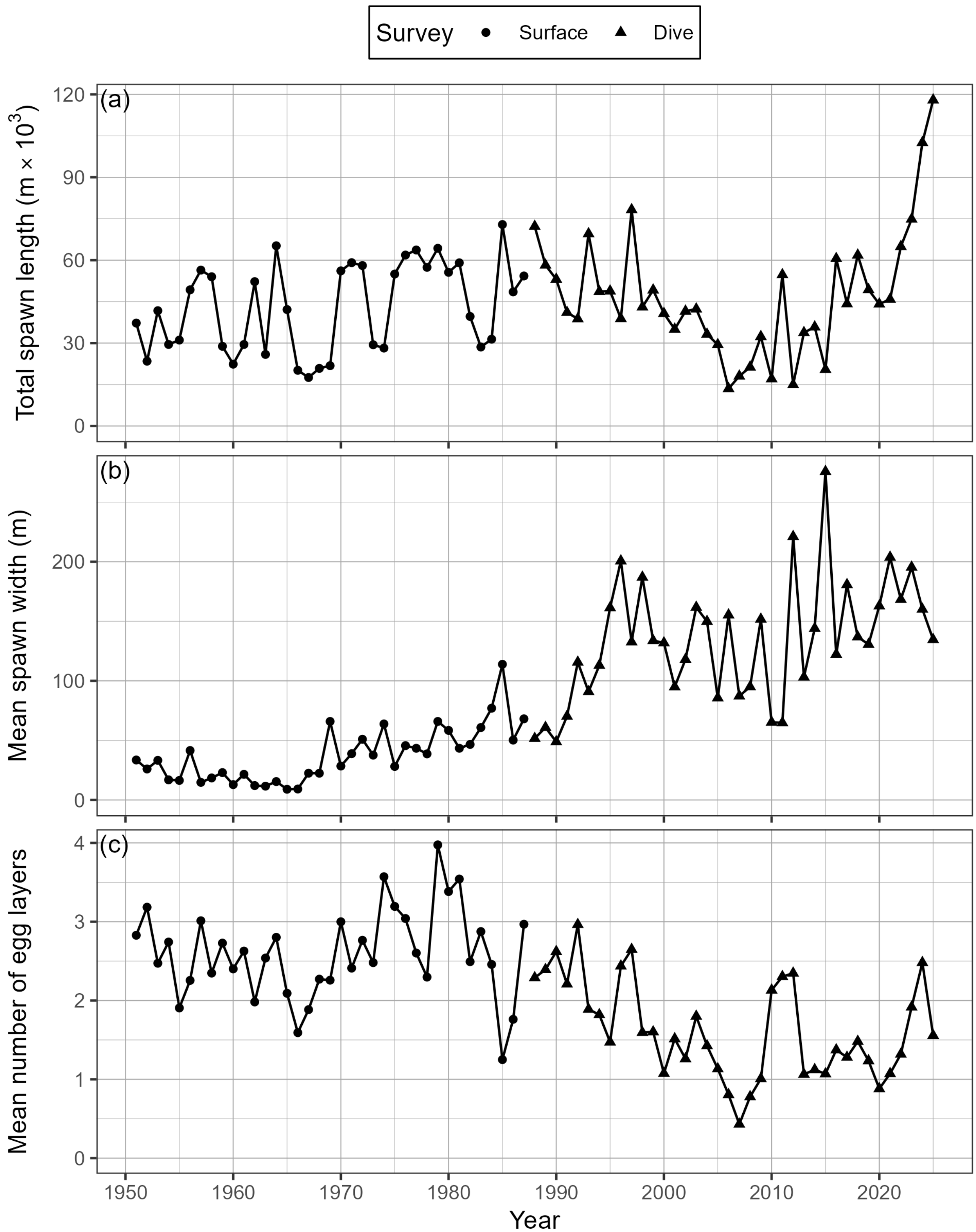


Figure 23. Time series of total spawn length in thousands of metres (m $\times$ 10^3; panel a), mean spawn width in metres (b), and mean number of egg layers (c) for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

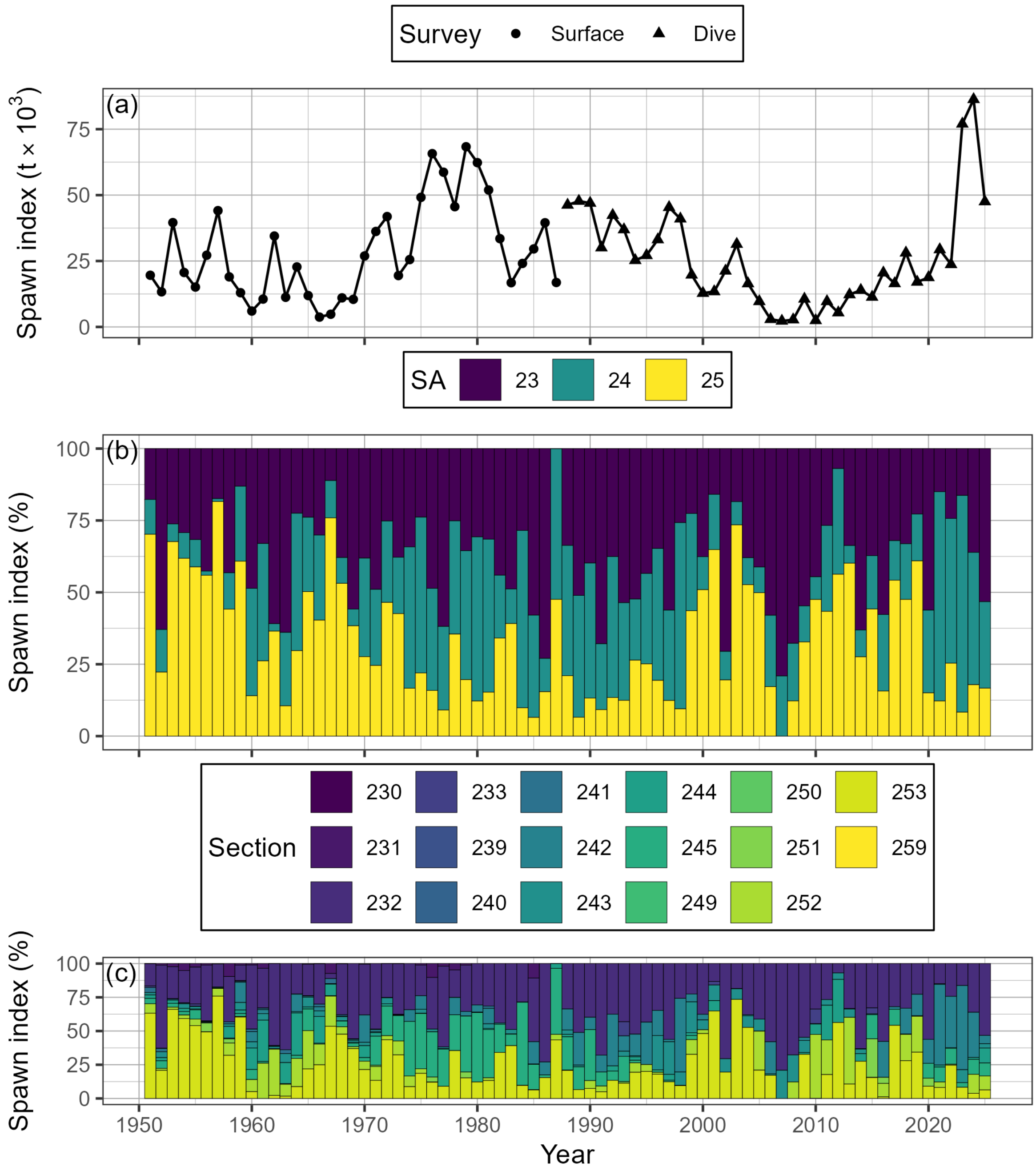


Figure 24. Time series of spawn index in thousands of metric tonnes (t $\times$ 10^3) for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR; panel a), as well as percent contributed by Statistical Area (SA), and Section (b, & c, respectively). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). Note that spawn surveys in the dive survey period (1988 to 2025) are a combination of surface and dive surveys (Figures 21 and 22). The 'spawn index' is not scaled by the spawn survey scaling parameter, q.

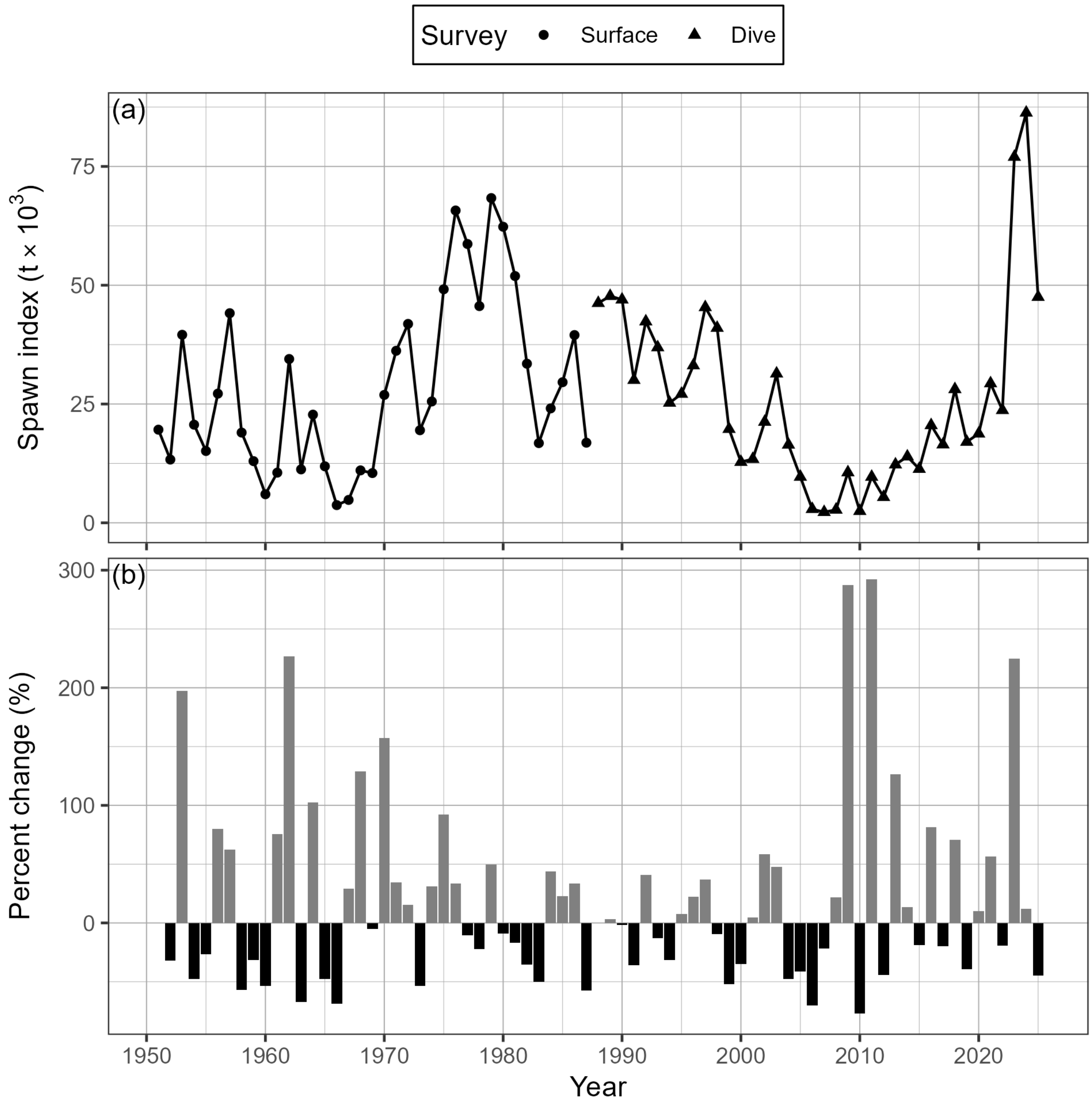


Figure 25. Time series of spawn index in thousands of metric tonnes (t $\times$ 10^3) for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR; panel a), and percent change (b). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the spawn index in year t. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Note that spawn surveys in the dive survey period (1988 to 2025) are a combination of surface and dive surveys (Figures 21 and 22).

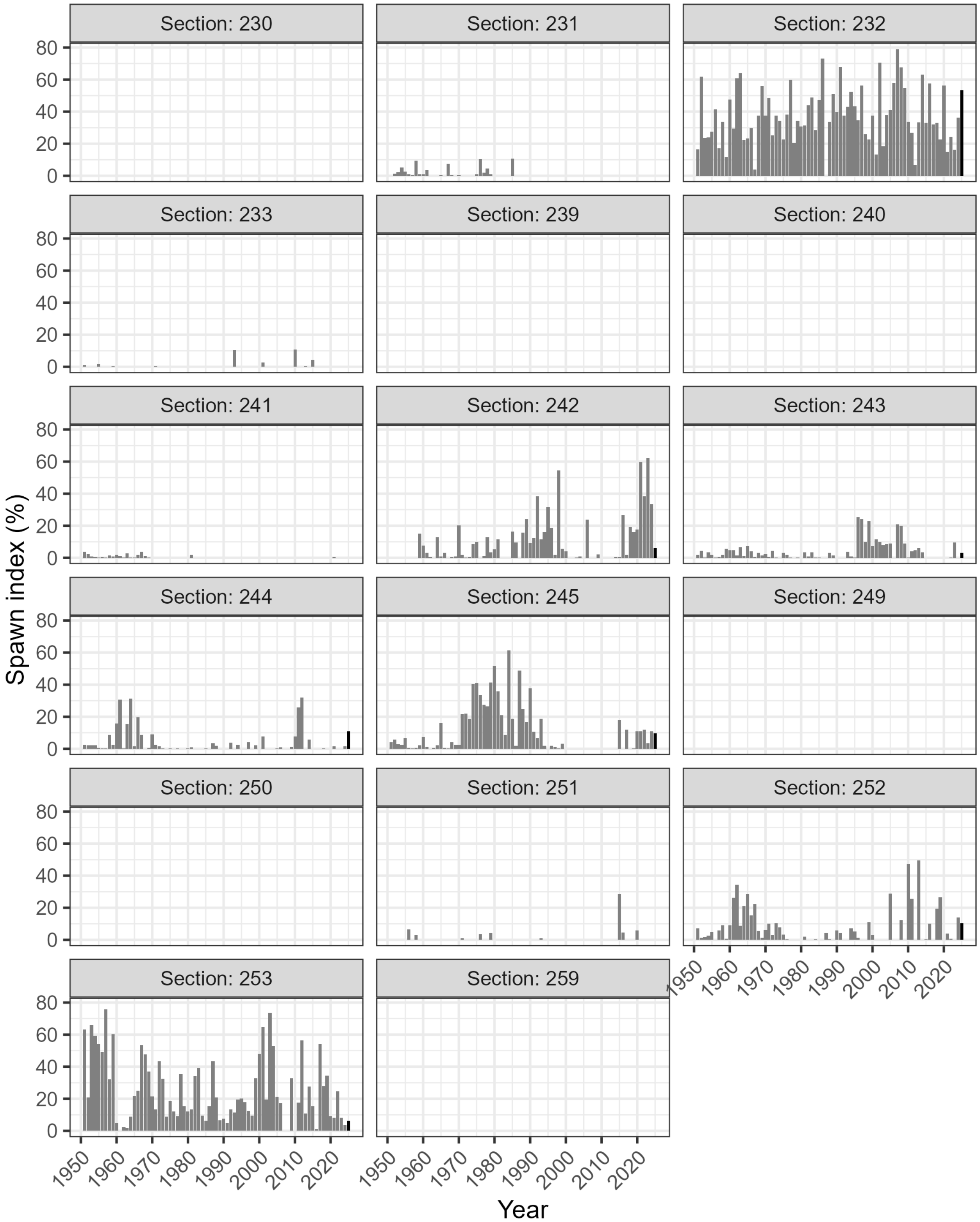


Figure 26. Time series of percent of spawn index by Section for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The year 2025 has a darker bar to facilitate interpretation. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The 'spawn index' is not scaled by the spawn survey scaling parameter, q.

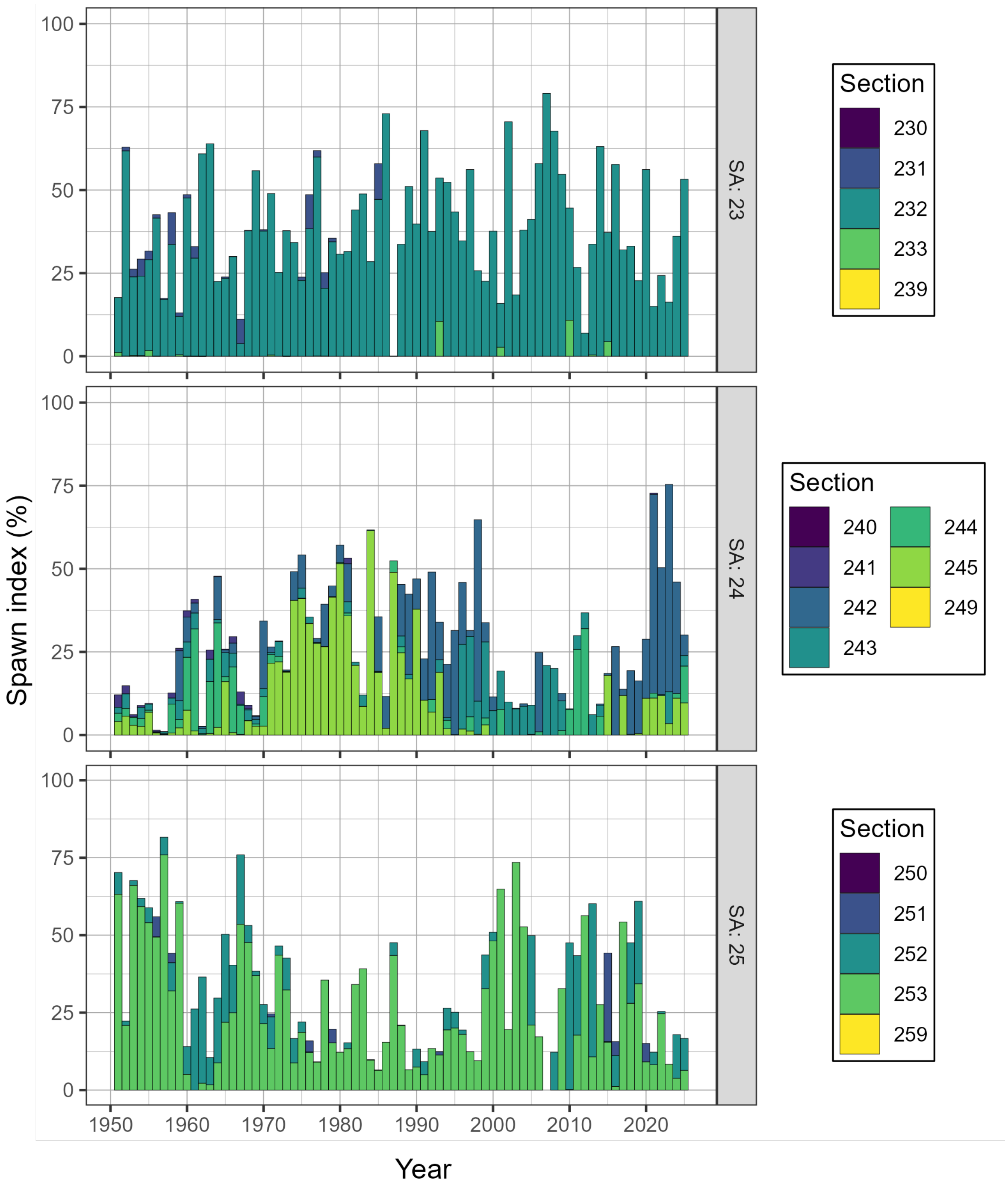


Figure 27. Time series of percent of spawn index by Statistical Area (SA) and Section for Pacific Herring from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The 'spawn index' is not scaled by the spawn survey scaling parameter, q.

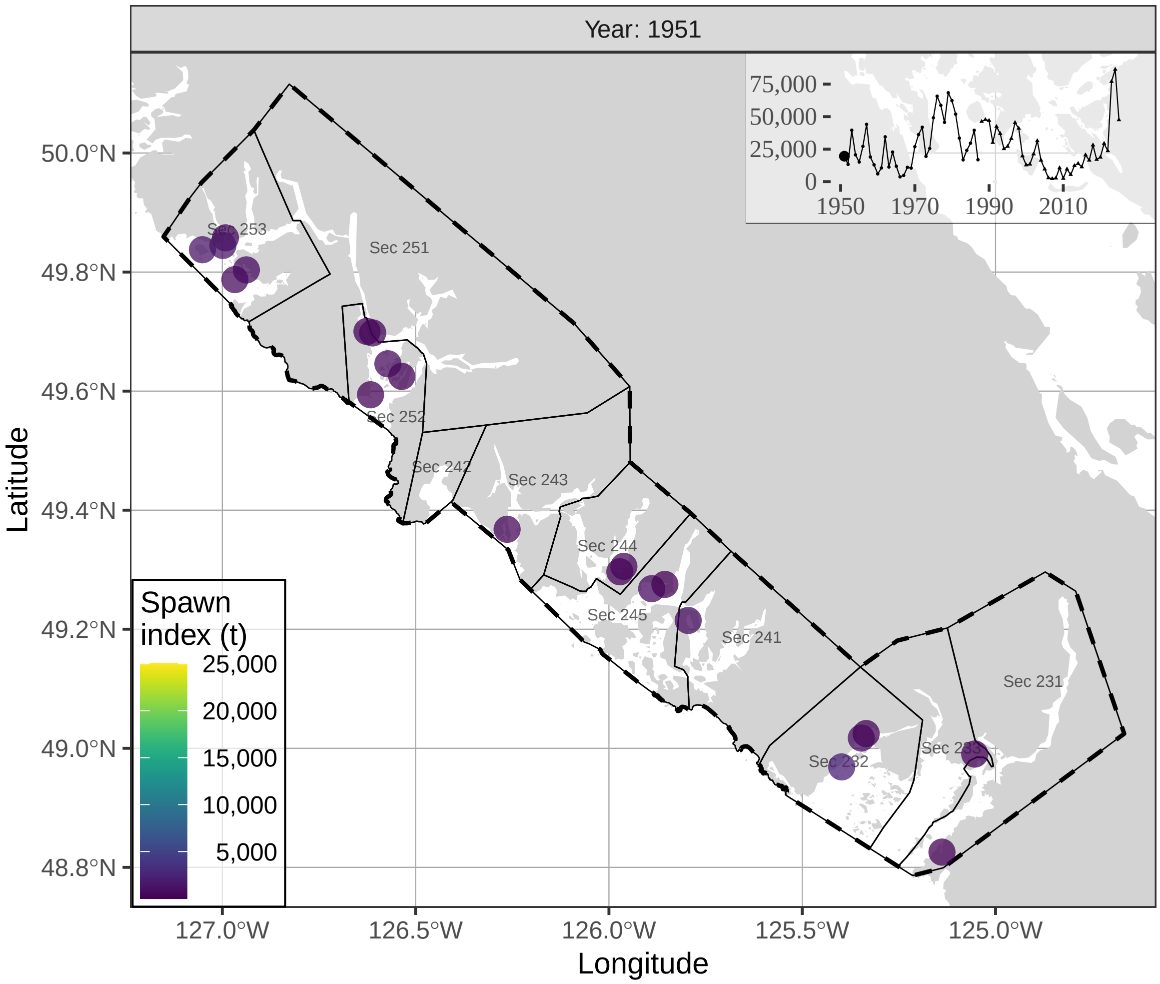


Figure 28. Animation of Pacific Herring spawn index in metric tonnes (t) by Location from 1951 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km). View the animation: download the report, open with Adobe, enable Java, and click "play".